



Investigating the impact of hospital antibiotic usage on aquatic environment and aquaculture systems: A molecular study of quinolone resistance in *Escherichia coli*

Sneha Kalasseri Girijan^a, Robin Paul^c, Rejish Kumar V.J.^{a,b}, Devika Pillai^{a,*}

^a Department of Aquatic Animal Health Management, Kerala University of Fisheries and Ocean Studies, Kochi, Kerala, India

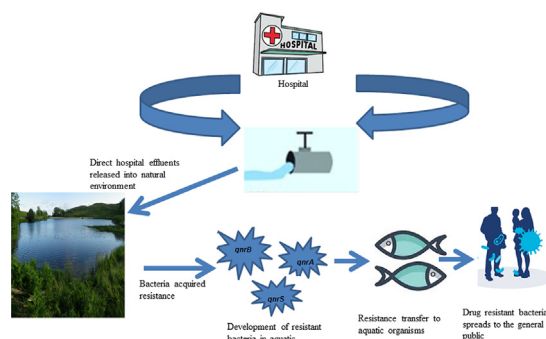
^b Department of Aquaculture, Kerala University of Fisheries and Ocean Studies, Kochi, Kerala, India

^c State Laboratory for Livestock, Marine & Agri Products (SLMAP), Department of Animal Husbandry, Government of Kerala, India

HIGHLIGHTS

- High quinolone resistance observed in *E. coli* isolates from Indian hospital wastewaters.
- Plasmid Mediated Quinolone Resistance (PMQR) and efflux pump genes were detected.
- *qnrB* was the most prevalent among the identified PMQR genes.
- Horizontal transfer of resistance gene was confirmed by conjugation.
- Antibiotic dissemination from direct hospital effluents to public waters was observed.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 13 June 2020

Received in revised form 4 August 2020

Accepted 4 August 2020

Available online 6 August 2020

Editor: Frederic Coulon

Keywords:

Antimicrobial resistance

Aquatic environment

Escherichia coli

Quinolone resistance

Multidrug efflux pump

Hospital effluents

ABSTRACT

Quinolones are one of the most important classes of antibacterials available for the treatment of infectious diseases in humans. However, there is a growing concern about bacterial resistance to antimicrobials including quinolones. The spread of antibiotic-resistant bacteria in the aquatic environment has been recognized as a growing threat to public health and hospitals appear to be a major contributor to this. The objective of this study was to investigate the prevalence of quinolone resistance in *Escherichia coli* from selected water bodies receiving direct hospital effluents in Kerala, India. Standard disc diffusion and E-test were used for antibiotic susceptibility testing. As antibiotic resistance can develop in bacterial isolates by different means, EtBr Agar Cartwheel method was used to detect the efflux pump activity and presence of resistant genes was detected by PCR. The mechanism of transfer of plasmid mediated resistance was confirmed by conjugation experiments. A total of 209 multidrug-resistant *Escherichia coli* were isolated from different hospital effluent discharge sites and aquaculture farms located in their vicinity. Among them, *qnrB* was found to be most prevalent followed by *qnrS*, *OqxAB*, *qnrA* and *aac (6')-Ib-cr*. The results suggested that the antibiotics present at sub-inhibitory concentrations in direct hospital effluents increases the selection pressure impacting the cell function of even normal microorganisms in the aquatic environment to change the genetic expression of virulence factors or acquire resistance genes by different transfer mechanisms, posing a serious threat to public health.

© 2020 Published by Elsevier B.V.

* Corresponding author at: Department of Aquatic Animal Health Management, Kerala University of Fisheries and Ocean Studies, Kochi, Kerala, India.

E-mail address: utpanangad@kufos.ac.in (D. Pillai).

1. Introduction

Antimicrobial agents are critical tools for fighting diseases in humans, domestic and aquaculture animals, but they are now becoming ineffective. Alarming levels of resistance have been reported in countries with many common diseases becoming untreatable and lifesaving medical procedures riskier to perform. Misuse and overuse of existing antimicrobials in humans, animals and aquaculture is accelerating the development and spread of antimicrobial resistance (AMR) coupled with no discovery of new antibiotics, and hence AMR is one of WHO's top 10 threats to global health (WHO, 2019). In India, one of the major drivers for AMR is the antibiotic use for human health and problems with sanitation, especially in view of rapid urbanization and privatization of health care and there are very few studies, which have attempted to investigate and understand the major environmental relations. The spread of antibiotic resistant bacteria (ARB) in the aquatic environment has been recognized as a growing threat to public health. Hospitals are a major contributor to the release and spread of ARB in the environment. Studies revealed that hospital discharge is a highly selective environment, consisting of different types of bacteria, nutrients and also antimicrobial agents, and provide a close contact between bacteria and antibiotics (Hirsch et al., 1999). It is a cause for concern because hospital effluent is generally discharged untreated into the main wastewater system and eventually into the environment. Many studies have been reported on the release and the prevalence of ARB in sewage and natural environments (Tacao et al., 2012). Studying the effect of effluents from hospitals and wastewater treatment plants (WWTP) on the receiving aquatic ecosystems, Proia et al. (2018) and Azuma et al. (2019) reported that the release of untreated or inadequately treated hospital effluent may potentially accelerate the spread of ARB and antibiotic resistant genes (ARGs). In addition, variety of other activities such as increasing population density along the river, industrial activities, agricultural and/or aquaculture activities (Jiang et al., 2017a) may contribute to the increase and spread of ARB. They can be transferred quickly into aquatic reservoirs and accumulated readily in sediments. The antibiotic residues increase the selection pressure; therefore the normal microorganisms in the aquatic environment acquire resistance through different types of transfer mechanisms like conjugation, transformation, and transduction (Marti et al., 2014). Indeed, in such specific environments, horizontal gene transfer would be favored, preferably directly, subject to anthropogenic impacts.

Empirical evidence suggests a close correlation between antibiotic use and the subsequent development of individual and community-level bacterial resistance (Bell et al., 2014). Van Boeckel et al. (2014) reported that India was the largest antibiotic consumer with 12.9/10⁹ units sold in 2010 (10.7 units per person). India accounted for 3% of the global consumption of antimicrobials in food animals in 2010 and was the fourth highest in the world, behind China (23%), the United States (13%) and Brazil (9%). The consumption of antimicrobials in the food animals sector in India is expected to double by 2030 (Van Boeckel et al., 2015). Antibiotic consumption per capita in India's retail sector is reported to have increased by around 22% over a five-year period from 2012 to 2016 (Farooqui et al., 2019). Klein et al. (2018) indicated that the consumption of antibiotics in India increased by 103%, from 3.2 to 6.5 billion Defined Daily Dose (DDDs) between 2000 and 2015, while the rate of consumption of antibiotics increased by 63% (from 8.2 to 13.6 DDDs). In a study conducted in Delhi, it was found that fluoroquinolones and cephalosporins were the most widely used antibiotics (Kotwani and Holloway, 2014).

Quinolones are one of the most commonly prescribed and widely used antibacterial agents to treat both community and hospital acquired infections. They are a group of antibiotics containing a bicyclic core structure related to the compound 4-quinolone. Based on the spectrum of activity, quinolones are classified into four generations. The first generation quinolone activity (e.g. nalidixic acid) was limited to Gram-negative organisms. The discovery that a key modification of

adding a fluorine (F) atom at the R6 position, could significantly improve the spectrum of activity (Pharma et al., 2009) led to the use of the second-generation quinolones. Ciprofloxacin, enoxacin, lomefloxacin, norfloxacin, and ofloxacin are second-generation agents. In addition to the fluoro substituent, these drugs were further modified by addition of a piperazine ring to the R7 position, a cyclopropyl group to the R1 position and an -OCH₃ substituent to the R8 position. The R7 piperazine ring improved the Gram-negative potency (Caekenberghe and Pattyn, 1984), while the cyclopropyl group was found to improve the overall activity of the compounds (Peterson, 2001). The third generation included addition of alkylated piperazine and pyrrolidinyl groups to the R7 position, and -NH₂, -OH, and -CH₃ groups to the R5 position to the pharmacophore. The R1 and R8 positions were kept unchanged from the second generation. The third generation also added new substituents such as chloro group at the R8 position. These modifications enabled an expanded activity against Gram-positive organisms (Ledoussal et al., 1999). Gatifloxacin, grepafloxacin, clinafloxacin and sparfloxacin are examples of third generation quinolones. Fourth generation quinolones covers all the activities of third generation drugs and extra anaerobic activity (Naber and Adam, 1998). The presence of nitrogen at the R8 position, azabicyclic group and a bulky side chain on the pyrrolidine group at the R7 position and addition of a difluoromethyl ether group at the R8 position, which improves the Gram-positive activity. Moxifloxacin, gemifloxacin, and trovafloxacin are the examples of fourth generation compounds. Among the quinolones, fluoroquinolones (FQs) are one of the most extensively prescribed and used antibiotics in hospitals, particularly for urinary tract and intestinal infections (Farooqui et al., 2019). These drugs were expected to bypass the problem of resistant bacteria, since no resistance mechanism specific for this class of antibiotics was supposed to exist in nature (Robicsek et al., 2006a). However, intensive clinical and agricultural use of FQs led to increased resistance levels among pathogenic microbes to these agents (Yamane et al., 2007). Resistance to this class of antibiotics has been commonly reported in clinical environments, municipal wastewater treatment plants, hospital wastewaters and urban waters following the introduction of quinolones into clinical practice (Martinez-Martinez et al., 1998; Novo and Manaia, 2010; Vredenburg et al., 2013).

In Enterobacteriaceae, quinolone resistance has emerged several times independently and is associated with chromosome mutations or plasmid-borne genes (Strahilevitz et al., 2009). Accumulations of mutations in the bacterial enzymes, DNA gyrase and DNA topoisomerase IV, targeted by fluoroquinolones are the primary mechanism of quinolone resistance, thus preventing bacterial DNA replication (Hooper, 2001). Such mutations can result in conformational changes in the enzymes, thus preventing the binding of quinolones to the DNA-substrate complex. In addition, plasmid-mediated quinolone resistance genes (PMQR) have also been described (*qnrA*, *qnrB*, *qnrS*, *qnrC* and *qnrD*), however; these genes possess only low-level resistance to quinolones. It has been suggested that their presence may favor the occurrence of chromosomal mutations, which will increase the tolerated minimum inhibitory concentration (MIC) and hence the level of resistance (Cavaco et al., 2009; Hata et al., 2005; Park et al., 2006; Wang et al., 2009). Another transferable quinolone resistance determinant, *aac* (6')-Ib-cr, possess an N-acetylating activity of the piperazinyl substituent of ciprofloxacin and norfloxacin. This enzyme, reported to be geographically wide-distributed, has been further identified to contribute to quinolone resistance (Park et al., 2006; Robicsek et al., 2006a; Robicsek et al., 2006b). Changes in the expression of efflux pumps and porin proteins are also a common mechanism for bacterial resistance to fluoroquinolones. For instance, *qepA*, an efflux pump belonging to the main facilitator subfamily (Yamane et al., 2007) and *OqxAB*, a multi-drug efflux pump that provide multiple agent resistance, have also been reported to reduce ciprofloxacin and nalidixic acid susceptibility (Hansen et al., 2007).

Escherichia coli are commensals in the intestinal flora of humans and warm-blooded animals. Many genotypes have specific virulence factors

and can cause disease such as acute gastroenteritis, urinary tract infections (UTIs), and sepsis/meningitis (Castillo et al., 2013). *E. coli* presence in water is commonly used as a fecal pollution and water quality microbiological indicator due to its anthropogenic origin (WHO, 2006). Members of this species harbor a variety of plasmids, considered to be correlated with the ability to acquire resistance genes from different families (Mammeri et al., 2005; Shibl et al., 2012). The existence of pathogenic *E. coli* in environmental water generates a potential danger for humans and animals; particularly as the same water is a source of drinking water, and is used for irrigation, and recreational purposes (Kummerer, 2009). *E. coli* has been associated with genes of antibiotic resistance that can be transferred to human and animal flora where they can exhibit intense pressure for the spread of resistance (Adefisoye and Okoh, 2016). In India, *E. coli* ($n = 1815$) isolated from the community showed high overall resistance to ampicillin, naladixic acid, and co-trimoxazole (75%, 73%, and 59%, respectively) between 2004 and 2007 (Holloway et al., 2009). Nearly a third of isolates were resistant to injectables like aminoglycosides (represented by gentamicin). From 2008 to 2013, *E. coli* resistance to third-generation cephalosporins increased from 70% to 83%, and fluoroquinolone resistance increased from 78% to 85% (CDDEP, 2015). There is not much data on the extent of resistance, with the exception of a few single-hospital reports, despite the need for robust national data in order to drive policy. A national surveillance platform is being built under ResistanceMap (www.resistancemap.org), a global data repository for antimicrobial use and resistance, which relies on reports from accredited laboratory service providers (CDDEP, 2015).

The objective of this study was to investigate the prevalence of quinolone resistance in *E. coli* from selected water bodies receiving direct hospital effluents and aquaculture farms in their vicinity in the Indian state of Kerala. In particular, the study aimed to identify the major quinolone resistance genes in *E. coli* isolates and the mode of transfer of plasmid mediated quinolone resistance by conjugation experiment.

2. Material and methods

2.1. Study area

Kerala is the southern coastal state with a long coastline and numerous rivers and associated inland water bodies. Water bodies adjacent to prominent hospitals located in three districts (Ernakulam, Kollam and Kannur) of Kerala, India (H1, H2 and H3) and five aquaculture farms (F1-F5) in their vicinity were selected for the study (Fig. 1). H1 is a multi-super specialty hospital with 350 patient beds, H2 is a government medical college with 500 patient beds and H3 is with 250 patient beds. All the three hospitals discharged their effluents directly into the water bodies where the water is shared and used for different anthropogenic purposes. There were no pharmaceutical industries adjacent to the water bodies.

2.2. Sample collection and processing

Sediment, water and fish/shrimp/clams were collected from the sampling points following standard procedures. Two sampling sites were selected (i) the point of exit of hospital effluents into public water bodies (ii) farms located downstream from the hospital sites. Water samples (1250 mL) from each site were collected in amber-coloured sterile bottles from outlet pipes of the hospitals, labeled H1, H2 and H3 and transported on ice to the microbiology laboratory within 2 h of collection; one litre of the sample was kept apart for antibiotic residue detection and 250 mL used for microbiological analysis. Approximately 100–200 g of sediment was taken from each site. For animal samples, 50 g of tissue was taken and homogenized. Samples were serially diluted and a volume of 100 μ L from each sample was inoculated onto the Trypticase soy agar (Himedia, India) and incubated at 37 °C

for 24 h. Following incubation, colonies were enumerated and the colonies with same morphology were randomly selected.

2.3. Microbiological analysis of suspected *E. coli* from samples

Total colony-forming units per plate within the range of 30–300 were included for descriptive analysis. The colonies were randomly selected and transferred at least three times to ensure its purity. Single colonies were then transferred to MacConkey agar (Himedia, India) and incubated for 24–72 h at ambient temperature until visible colonies were observed. Isolates that appeared bright pink on the selective agar were confirmed as *E. coli* and further inoculated into Trypticase soy broth for biochemical characterization for species confirmation (Holt et al., 1994).

2.4. Antimicrobial susceptibility testing

Antibiotic resistance phenotypes were determined based on the disc diffusion method (Bauer et al., 1966), according to the guidelines of CLSI (Clinical and Laboratory Standards Institute, 2013). All the antibiotic discs and the media were purchased from Himedia, India. Overnight cultures of the bacterial isolates were prepared in liquid media and adjusted to 0.5 of a McFarland standard and plated on Muller-Hinton agar (Himedia, India). The following antibiotic discs were used for antimicrobial susceptibility testing: ciprofloxacin (5 μ g), levofloxacin (5 μ g), moxifloxacin (5 μ g), ofloxacin (5 μ g), norfloxacin (10 μ g), nalidixic acid (30 μ g), gentamicin (10 μ g), amikacin (30 μ g), cefotaxime (30 μ g), cefoxitin (30 μ g), cefotetan (30 μ g), ceftazidime (30 μ g), imipenem (10 μ g) and colistin (10 μ g) followed by incubation at 37 °C for 24 h. The minimum inhibitory concentration (MIC) was determined for selected isolates using strips embedded with an antibiotic gradient (Himedia, India).

2.5. Detection of efflux system by Ethidium bromide (EtBr)-agar Cartwheel (EtBrCW) method

All the *E. coli* isolates with resistance towards ciprofloxacin by disc diffusion method were selected for efflux pump analysis. This is a simple, non-instrumental, agar based method that uses EtBr to demonstrate efflux pump activity in bacteria (Martins et al., 2011). It provides information on the ability of each isolate to extrude EtBr from the cells by efflux on the basis of fluorescence emitted from isolates swabbed in EtBr-containing agar plates. The technique employed was to use different sets of freshly prepared plates of Trypticase Soy Agar (TSA) with concentrations of EtBr ranging from 0.5 to 2.5 mg/L and kept protected from light until use. Overnight cultures of the bacterial isolates were prepared in Luria-Bertani broth and adjusted to 0.5 of a McFarland standard. Then, from the centre of the plate to the margin, the bacterial cultures were swabbed on the EtBr-TSA plates and incubated for 16 h at 37 °C. After incubation, the TSA plates were examined under gel-imaging system. Higher the concentration of ethidium bromide required to produce fluorescence of the bacterial mass, greater the efflux capacity of the bacterial cells.

2.6. Molecular detection of quinolone and efflux pump encoding genes

All the plasmid-mediated quinolone genes (*qnrA*, *qnrB*, *qnrS*, *qnrC* and *qnrD*), *aac* (6')-Ib-cr, *qepA*, *OqxAB* and *qac* Δ 1 were detected as described by (Chen et al., 2012; Jia et al., 2015; Jiang et al., 2017b; Park et al., 2006; Robicsek et al., 2006c). Details of the primers used are given in Table 1. The PCR reaction was performed in a final volume of 25 μ L containing 0.125 U Taq DNA polymerase, 1 $\times$ PCR buffer, 1.5 mM MgCl₂, 0.2 mM dNTPs and 0.5 μ M of primers. All the PCR products were purified and sequencing was performed with an automated ABI 3100 Genetic analyzer using ABI BigDYE terminator method (M/s

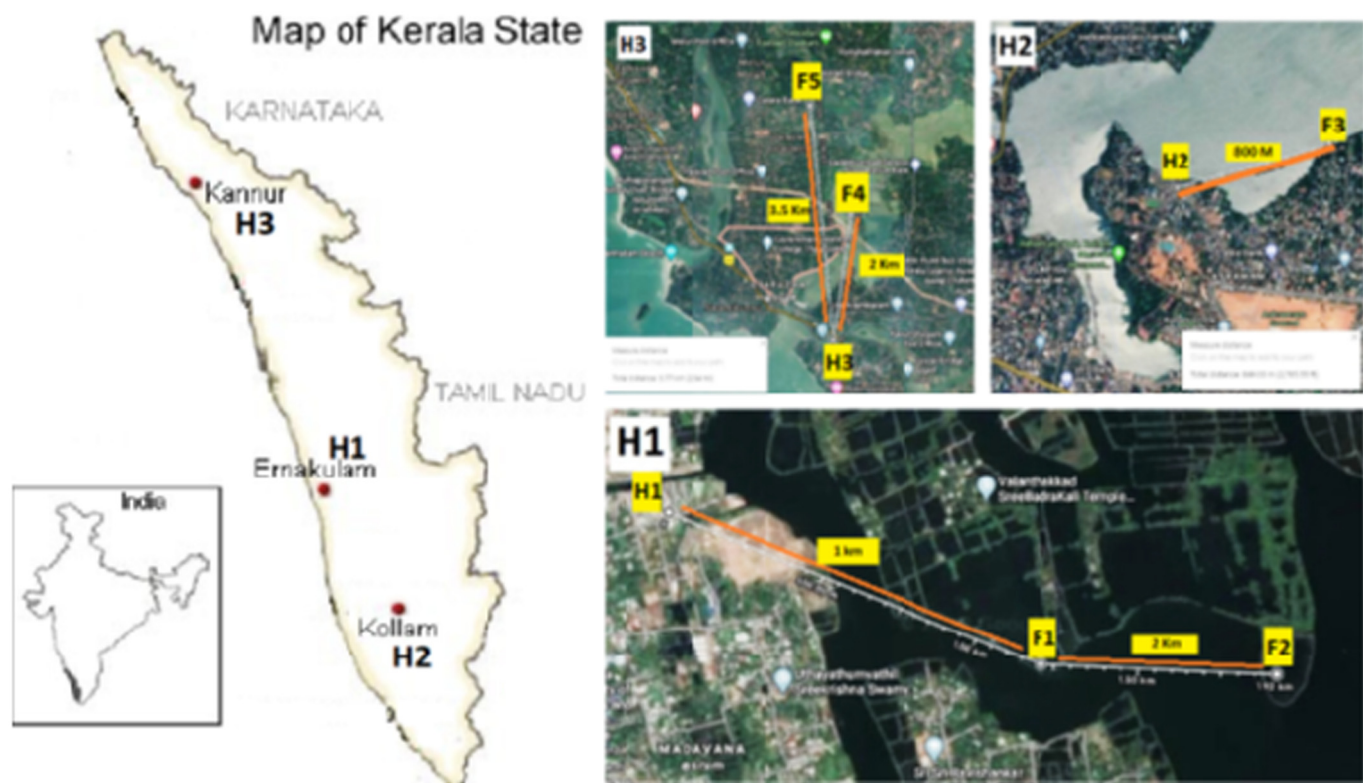


Fig. 1. Sample collection sites. The arrow denotes the distance between hospital and aquaculture farms.

Agrigenome Pvt. Ltd., Kochi). The nucleotide sequences were analyzed using the BLAST algorithm (<https://www.ncbi.nlm.nih.gov/BLAST>).

2.7. Conjugation study

Conjugation experiments were carried out by a broth mating method as described earlier (Wang et al., 2008) using azide resistant (AzR) *E. coli* J53 as recipient and *qnrS* positive *E. coli* as the donor. *E.*

coli J53AzR was kindly gifted by Dr. Sanath Kumar (Central Institute of Fisheries Education, Mumbai, India). Pure colonies of *qnrS* positive *E. coli* and recipient cells were inoculated separately and incubated overnight at 37 °C with shaking. These overnight cultures were diluted 1:100 in fresh medium, and each was grown to early exponential phase. Mating mixture was prepared by adding 0.1 mL of donor cells to 0.9 mL of recipient cells and was gently swirled for a few minutes and then incubated at 37 °C for 6 h (without shaking) followed by

Table 1
List of primers used for PCR amplification.

Specific gene for amplification	Primer	Primer sequence (5'-3')	Amplicon size	Reference
<i>qnrA</i>	Forward	ATTTCACGCCAGGATTG	516 bp	Robicsek et al., 2006c
	Reverse	GATCGGCAAAGTTAGGTCA		
<i>qnrB</i>	Forward	GATCGTGAAAGCCAGAAAGG	469 bp	Robicsek et al., 2006c
	Reverse	ACGATGCCTGGTAGTTGTC		
<i>qnrS</i>	Forward	ACGACATTGCTCAACTGCAA	417 bp	Robicsek et al., 2006c
	Reverse	TAAATTGGCACCCTGTAGGC		
<i>qnrC</i>	Forward	GGGTGTACATTTATTGAATC	447 bp	Chen et al., 2012
	Reverse	TCCACTTTACGAGTTCT		
<i>qnrD</i>	Forward	GCAAGTTCATTGAACAGGCT	428 bp	Chen et al., 2012
	Reverse	TCTAAACCGTCGAGTTCGGCG		
<i>aac (6')-Ib-cr</i>	Forward	TTGCGATGCTCTATGAGTGGCTA	482 bp	Park et al., 2006
	Reverse	CTCGAATGCTGGCGTGTTT		
<i>qepA</i>	Forward	CTGCAGGTAAGCGTCATG	403 bp	Chen et al., 2012
	Reverse	CGTGTGCTGGAGTCTCTC		
<i>OqxA</i>	Forward	CTTGCACTTAGTTAAGCGCC	866 bp	Jia et al., 2015
	Reverse	GAGGTTTGTAGTGGAGGTAGG		
<i>OqxB</i>	Forward	GCGGTGCTGTCGATTTA	781 bp	Jia et al., 2015
	Reverse	TACCGGAACCCATCTCGAT		
<i>qac EΔ1</i>	Forward	AAGTAATCGCAACATCCG	140 bp	Jiang et al., 2017a, 2017b
	Reverse	ATAAGCAACACCGACAGG		
<i>bla_{CTX-M}</i>	Forward	CGCTTTGCGATGTGCAG	550 bp	Villegas et al., 2004
	Reverse	ACCGCATATCGTTGGT3		
<i>bla_{SHV}</i>	Forward	TTAATCCCTGTTAGCCA	795 bp	Sharma et al., 2010
	Reverse	GATTTGCTGATTTCGCC		

plating on Luria agar medium (Himedia, India) containing sodium azide (100 µg/mL from Sigma Aldrich, United States) for counter selection and ampicillin (100 µg/mL) for plasmid-encoded resistance. The colonies were replica-plated on TSA agar plates supplemented with and without ciprofloxacin to ensure that quinolone resistance was co-transferred. Transconjugants carrying the same *qnrS* gene as their donors were verified by PCR.

2.8. Antibiotic residue detection

50 mL of water sample was filtered through 0.45 µm membrane filter; the sample was acidified with 1 N H₂SO₄ to pH 3 and loaded on C-18 activated cartridge (The residue was reconstituted with acetonitrile to make a final volume of 1 mL). Sediment samples were also homogenized well before analysis. 100 mL of acidified water (pH 3.5 with phosphoric acid) was added to 20 g of samples, mixed with shaking for 1 h and filtered using Whatman filter paper.

Ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS) was used for antibiotic residue detection, following the procedure by Saxena et al. (2018) with modifications. The chromatographic analysis was performed using Acquity ultra high performance liquid chromatography system (Waters, Milford, MA, USA). The chromatographic separation was carried out using an XBridge BEH C18, 2.5 µm, 2.1 × 100 mm (Waters) with mobile phase consisting of Methanol (phase A) and 0.1% formic acid in water (phase B) at a flow rate of 0.5 mL.min⁻¹. Mass spectrometry analysis was performed using Xevo TQ-D triple quadrupole mass spectrometer (Waters Corp., Ireland) with an electrospray ionization (ESI) interface, capillary voltage was 3.0 kV in positive mode. Each target compound was detected by using two multiple reaction monitoring (MRM) channels. Of the two transitions, one was used for quantification and the other for confirmation. Instrument settings, data acquisition and processing were controlled by the software Masslynx (version 4.1, Waters). Ciprofloxacin, Norfloxacin, levofloxacin, nalidixic acid and sulfamethoxazole were analyzed in this study.

3. Results

3.1. Bacterial identification and antibiotic susceptibility test

A total of 524 Gram-negative bacilli were isolated from the study area, of which 366 isolates (59%) showed multi-drug resistance (MDR) to different class of antibiotics tested. A total of 135 *E. coli* isolates were identified from various samples screened from direct hospital effluents (H1 = 40, H2 = 67, H3 = 28). In comparison, a total of 74 *E. coli* isolates were obtained from aquaculture farms downstream of the above sites (F1 = 16, F2 = 9, F3 = 31, F4 = 11, F5 = 7). The number of cultivable ARB in the H2 samples was much higher than in the H1 and H3 samples. Occurrence and distribution of *E. coli* from different locations is shown in Table 2. In all the three direct hospital effluents, a large number of *E. coli* isolates displayed high resistance to at least one of the classes of antibiotics tested, with higher resistance to quinolones and cephalosporin than to aminoglycosides. Of the total 135 *E. coli* isolates from hospital sites, maximum resistance was found against ciprofloxacin (67%) (H1 = 29, H2 = 51, H3 = 10); followed by ofloxacin (H1 = 24, H2 = 38, H3 = 4), norfloxacin (H1 = 23, H2 = 32, H3 = 3), levofloxacin (H1 = 24, H2 = 29, H3 = 3), nalidixic acid (H1 = 20, H2 = 28, H3 = 2) and moxifloxacin (H1 = 22, H2 = 26, H3 = 3). A total of 41 (30.3%) isolates (H1 = 17, H2 = 23, H3 = 1) showed combined resistance to cefoxitin and cefotaxime or ceftazidime and cefotetan (28.1%, H1 = 15, H2 = 22, H3 = 1). The prevalence of isolates resistant to aminoglycoside was very less; 11.1% of the isolates from H2 alone showed resistance to amikacin and gentamicin. In aquaculture farm sites too, ciprofloxacin resistant isolates were found to be higher. Quinolone and cephalosporin resistant isolates were highest in F3, the farm which was very closely located (800 m) to the hospital discharge

Table 2
Occurrence and distribution of *Escherichia coli* from different locations.

Sampling area	Location and distance from main discharge to aquaculture farms	Types of samples	Total Gram-negative isolates	Total Number of multi-drug isolates	Number of <i>E. coli</i> isolates
Ernakulam	Direct hospital effluent (H1)	Water Soil Fish Clam	114	85	40
Ernakulam	Aquaculture Farm 1 (F1-1Km)	Water Soil Fish Clam	55	36	16
Ernakulam	Aquaculture Farm 1 (F2-2Km)	Water Soil Fish Clam	36	27	9
Kollam	Direct hospital effluent (H2)	Water Soil Fish Clam	154	116	67
Kollam	Aquaculture farm 3 (F3-800 m)	Water Soil Fish Clam	79	45	31
Kannur	Direct hospital effluent (H3)	Water Soil Clam	63	49	28
Kannur	Aquaculture farm (F4-2Km)	Water Soil Shrimp	15	5	11
Kannur	Aquaculture farm (F5-3.5Km)	Water Soil Shrimp	8	3	7

site (H2), followed by F1 (1 km away from H1) and F2 (2 km away from H1). Out of the 74 isolates, 19 *E. coli* isolates from F3 showed resistance to ciprofloxacin and ofloxacin ($n = 19$), followed by levofloxacin ($n = 13$), moxifloxacin ($n = 11$), norfloxacin ($n = 10$) and nalidixic acid ($n = 10$). In contrast, 10 isolates from F1 and 3 isolates from F2 showed resistance to ciprofloxacin followed by ofloxacin, levofloxacin, moxifloxacin, norfloxacin, and nalidixic acid. Cephalosporin resistance among the isolates from different aquaculture sites was highest among F3 (16.2%) followed by F1 isolates (6.7%). All the F2 isolates were susceptible to cephalosporins. In contrast, *E. coli* isolates from F4 and F5, located at a distance of 2 km and 3.5 km from H3 site respectively, were susceptible to almost all antibiotics, with very few isolates showing resistance to nalidixic acid alone. All *E. coli* isolates from different locations were susceptible to imipenem and colistin. The antibiotic resistance pattern to selected antibiotics for the 209 *E. coli* isolates collected from direct hospital effluents and aquaculture farms is shown in Figs. 2 and 3.

In order to correlate the fluoroquinolone resistance in the isolates with *qnr* gene determinants, the MIC for all the quinolone-resistant isolates was determined. It was noticed that the MIC values of the fluoroquinolones varied among *E. coli* with *qnr* genes. Of the 122 ciprofloxacin-resistant *E. coli* strains studied from various locations, 110 were highly resistant to ciprofloxacin with a range of ciprofloxacin MIC of 8–64 µg/mL. The MICs for ciprofloxacin, cefotaxime, ceftazidime, cefepime, amikacin and gentamicin are shown in Table 3.

3.2. Efflux pump detection by EtBr-agar Cartwheel method

The efflux pump activity among 90 ciprofloxacin resistant isolates from direct hospital effluents and 32 isolates from different aquaculture farms was assessed. A range of fluorescent bacterial masses were detected after incubation, depending on their capacity to efflux EtBr (Fig. 1S shown in supplementary data). This method allowed the

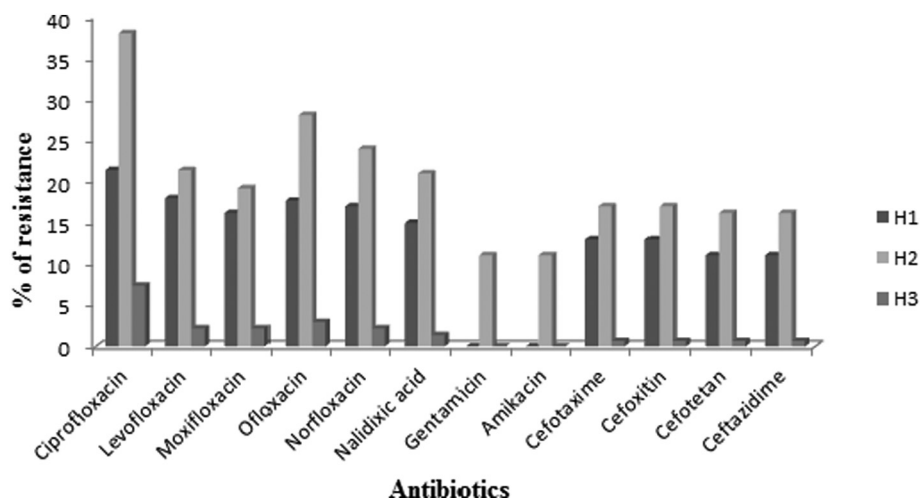


Fig. 2. Antibiotic resistance percentage of *Escherichia coli* isolated from different direct hospital effluents.

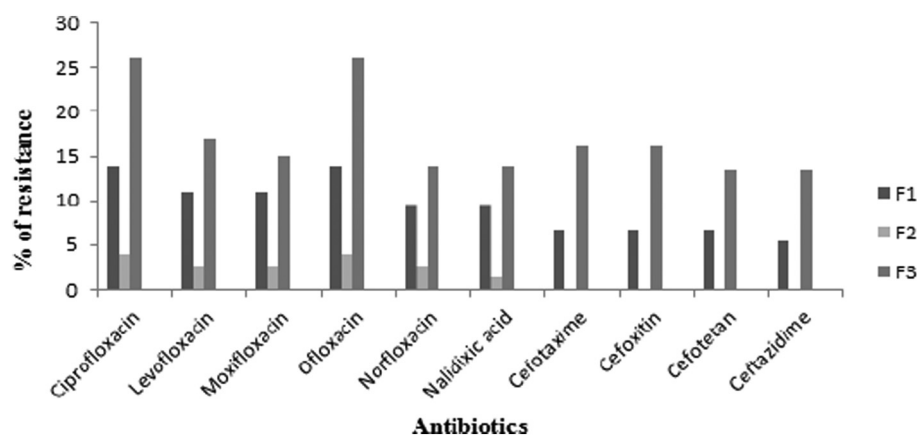


Fig. 3. Antibiotic resistance percentage of *Escherichia coli* isolated from different aquaculture farms.

selection of 27 *E. coli* isolates from H1 and H2 study area, 3 from H3 and 12 isolates from F3 showing increased EtBr efflux activity in plates with increasing concentrations of EtBr (2.5 mg/L). Whereas 8 isolates from H1 and 15 isolates from H2 that fluoresced at a concentration between 0.5 and 1.0 mg/L of EtBr concentration were considered as intermediate efflux active isolates, those isolates showing no fluorescence below 0.5 mg/L concentration of EtBr were designated as EtBrCW-negative isolates.

3.3. Molecular characterization of quinolone and efflux pump genes

We screened all 122 quinolone-resistant isolates from different study areas for each of the *qnr* genes associated with plasmids. Among

Table 3

MIC values for quinolone and cephalosporin resistant *Escherichia coli* isolates from different sampling sites.

Antibiotics	H1	H2	H3	F1	F2	F3
Ciprofloxacin	32	64	8–16	18	8	32
Ofloxacin	>64	128	>8	>32	2–8	64
Levofloxacin	16–32	128	8	8–16	4	64
moxifloxacin	32–64	128	16	16–32	4	64
Norfloxacin	32	128	8	32	2–4	64
Cefotaxime	>128	256	16–32	>32	16–32	128
Ceftazidime	128	128	16–32	32	16	>64
Cefoxitin	2–128	256	32	64	8–16	>64

MIC, minimum inhibitory concentration.

Tested range of antibiotics is 0.125–>512 mg/L^{−1}.

the PMQR-positive isolates, 11 isolates from H1, 19 from H2 and 2 from H3 showed the existence of *qnrB* specific PCR product, with the expected amplicon size of 469 bp (Fig. 4). Among the 32 *E. coli* isolates with quinolone resistance from aquaculture farms, 2 isolates from F1 and 6 from F3 showed the presence of *qnrB* gene. The number of isolates harbouring *qnrS* gene was 8 in H1 and 14 in H2 (Fig. 2S shown in supplementary data) and 3 isolates in F3. The sequence of the amplified product (GenBank accession number MT254541 and MT358339) showed close homology with *qnrB* and *qnrS* gene from other bacterial isolates. However, no isolate was found to be positive for *qnrS* gene in H3 area. Two isolates were found positive for *aac* (6′)-*lb-cr* and *qnrA* from H2. The gene *OqxAB* coding for plasmid-mediated efflux was detected in 6 isolates from H2 location (Fig. 3S shown in supplementary data) (GenBank accession number MT254542). Overall, *qnrB* and *qnrS* were the most prevalent PMQR genes followed by *OqxAB*, *aac* (6′)-*lb-cr* and *qnrA*. The genes *qnrC* and *qnrD* were found to be negative for all the *E. coli* isolates. The prevalence of the PMQR determinants among *E. coli* isolates of different locations is shown in Table 4.

3.4. Conjugation

The plasmid-mediated quinolone resistance was successfully transferred from *E. coli* with *qnrS* gene to azide-resistant *E. coli* J53 isolate. Plasmid isolated from the transconjugants and analyzed by PCR showed the specific amplification of *qnrS* gene in the transconjugants. Furthermore, the minimum inhibitory concentration of ciprofloxacin in the *E.*

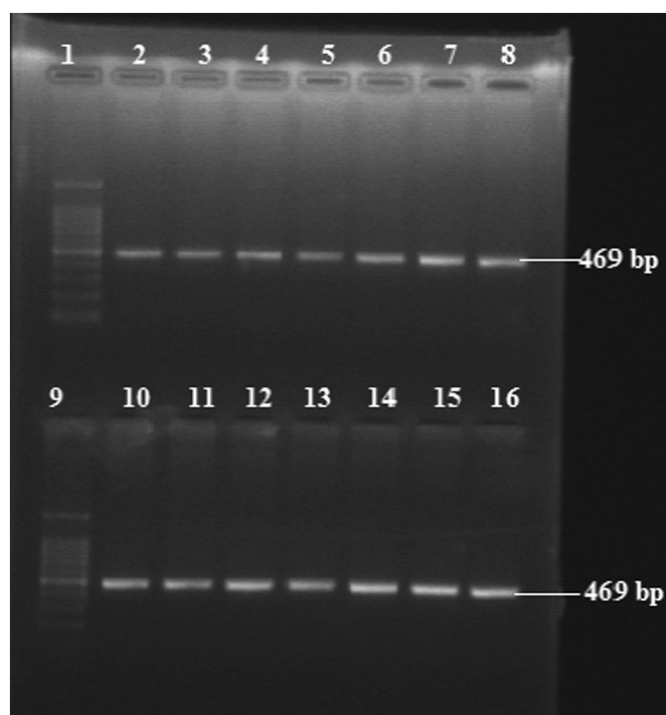


Fig. 4. Amplification of *qnrB* gene with 469 bp amplicon size product; lane 1: 100 bp molecular weight marker; lane 2: positive control. Lane 3–8: samples from Ernakulam (H1). Lane 9: 100 bp molecular weight marker; lane 10–16: samples from Kollam (H2).

Table 4

Prevalence of the PMQR encoding genes among *Escherichia coli* from different locations.

Sample location	ESBL Positive isolates	EtBr Positive isolates	<i>qnrB</i>	<i>qnrS</i>	<i>qnrA</i>	<i>aac</i> (6')-Ib-cr	<i>OqxAB</i>
H1	10	20	11	8	–	–	–
F1	2	–	2	–	–	–	–
F2	–	–	–	–	–	–	–
H2	15	30	19	14	2	2	6
F3	5	12	6	3	–	–	–
H3	–	3	2	–	–	–	–

coli transconjugants showed an MIC of 16 µg/mL whereas the strain without the plasmids was sensitive to ciprofloxacin.

3.5. Antibiotic residue detection

With an aim to find corroborating evidence of persisting residues of antibiotics in the environment, a total of 30 Multiple Reaction Monitoring (MRMs) were done for quantifying using UPLC–MS/MS for 5 different antibiotics from six environmental samples. Only one sediment sample collected from H2 sampling site showed the presence of ciprofloxacin. The detected level of ciprofloxacin residue was 15 ppb (Table 1S shown in supplementary data).

4. Discussion

Quinolones and especially fluoroquinolones are critically important class of broad spectrum antimicrobial drugs commonly used to treat Gram-negative bacterial pathogen infections. *E. coli* is recognized as one of the main cause of several extra-intestinal and hospital-acquired infections as it exhibits resistance to most quinolones including fluoroquinolones (Shetty et al., 2019).

Aquatic habitats such as rivers and streams are considered as ideal reservoirs for the transmission of antibiotic resistance, as antimicrobials and antimicrobial-resistant bacteria are often released directly into the ecosystem (Kummerer, 2009; Zhang et al., 2009) leading to serious contamination risks (Jang et al., 2013; Mazari-Hiriart et al., 2008). Szczepanowski et al. (2009) suggested that WWTPs from hospitals could be one of the routes leading to the spread of antibiotic-resistant bacteria into the environment. The presence of bacterial isolates with plasmid encoded resistance gene in WWTPs final effluents confirms that resistance determinants are released into the environment which might enhance further dissemination among environmental bacteria. Similarly, a study conducted by Haberecht et al. (2019) found that the total bacterial abundance detected in WWTP was 1.7 times higher than the sewer water. The water bodies with high proportion of multi-drug resistant Gram-negative isolates imply the release of effluents into the public waters without any proper management. This is of importance in the state of Kerala because the same water is used for irrigation and aquaculture purposes, apart from fishing activities, and can thus be a potential source of bacterial contamination that can harm humans or animals through direct contact, or the ingestion of contaminated food. The state has 44 rivers and an interconnected system of brackish water lakes and river estuaries that lies inland from the coast and runs virtually the length of the state. Most of the hospitals and industries are near thickly populated riversides, often near cities and towns. MDR isolates could, therefore, spread rapidly through these water bodies.

In the present study it was observed that high frequency of *E. coli* with multi-drug resistance isolated from the water bodies could be originating from a clinical setting as their numbers were much more in direct hospital effluent than in aquaculture farms. *E. coli* isolates with high antibiotic-resistance in public water bodies receiving direct hospital effluents could be due to the improper wastewater treatment, urine and fecal excretion from patients, direct disposal of expired medicines and accidental chemical or antibiotic spilling as suggested by Zulkeffle et al. (2016).

The “Hospital Size” has been observed as an important factor where bigger hospitals contributed greater bacterial loads to their surroundings relative to smaller hospitals (Lamba et al., 2017). A similar observation was made in the present study as well, with higher bacterial load as well as greater number of resistant isolates being observed in the discharge sites of bigger hospitals. More number of highly resistant *E. coli* isolates was found in H2, the hospital with the largest number of beds selected in the present study, as compared to H1. Least number of resistant isolates were found in H3, the smaller hospital in the study with only 250 bed capacity.

Antibiotic prescription levels for certain types of antibiotics in India are higher than for developed nations. For example, in India, the percentage of cephalosporin and quinolone prescriptions (38.2% and 16.3%) was substantially higher than in the USA (14.0% and 12.7%) and Greece (32.9% and 0.5%) (Kourilaba et al., 2015). According to WHO (2009) pilot study, the prescription percentage of antibiotics in Delhi, Vellore and Mumbai was 21.5%, 41% and 36–40% respectively. The most commonly used antibiotics were fluoroquinolones, cephalosporins, extended spectrum penicillins, macrolides, cotrimoxazole and tetracycline. It was found that quinolones such as ciprofloxacin, levofloxacin, and moxifloxacin possess immense market potential and sales value. In India, fluoroquinolones represent a 30% share of counterparts in the global pharmaceutical market. The usage of third generation cephalosporins that belong to the Watch category is increasingly growing. Three most commonly prescribed cephalosporins were cefuroxime, cefixime and combination of cefixime and clavulanic acid (Kotwani and Holloway, 2011). In comparison to the rapid increase in third-generation cephalosporin consumption between 2000 and 2015, fluoroquinolone consumption (Watch group antibiotics) is declining while penicillins (Access group antibiotics) consumption remains constant (Gandra and Kotwani, 2019). Analyzing the outpatient antibiotic

prescription rates in the private sector in India, Farooqui et al. (2019) reported that for all age groups, the prescription rates for cephalosporins was the highest (38.3%) followed by beta-lactam, penicillins (22.8%) and quinolones (16.3%).

Out of the total *E. coli* isolates obtained in the present study, 58.3% showed high resistance to quinolone class of antibiotics, particularly to ciprofloxacin. Among them, 29% displayed combined resistance to cephalosporins and a low percentage showed resistance to aminoglycosides (7.1%). A similar study from India revealed that the resistance to cephalosporins and quinolones was more frequent than to aminoglycosides and imipenem in *E. coli* isolated from hospital wastewaters (Chandran et al., 2014). High rates of quinolone resistance were observed in *E. coli* (63.8%) in hospital wastewaters in Tehran, Iran. The authors suggested that wastewater of hospitals is an important source of quinolone resistance (Ranjbar and Farahani, 2017). Extended-spectrum beta-lactamase (ESBL) producing *E. coli* with ciprofloxacin-resistance was isolated from Ireland's hospital effluent samples, signifying that these genes can coexist (Galvin et al., 2010). Co-selection of quinolone and cephalosporin resistance phenotypes was previously recorded in the clinical environment for Enterobacteriaceae and could probably lead to MDR for bacteria (Lavilla et al., 2008; Poirel et al., 2006). Earlier research on PMQR genes showed that their levels are significantly correlated with other genes for antimicrobial resistance, in particular ESBL genes (Wu et al., 2007; Yang et al., 2008; Wu et al., 2016) and both the PMQR and ESBL genes are regulated by the same promoter (Ma et al., 2009; Xia et al., 2013). This leads to the co-existence of multi-drug resistance mechanisms conferring resistance to many other antibiotic classes (Gniadkowski, 2001). A similar type of resistance pattern was observed in *E. coli* isolated from hospital wastewater in Central India where they exhibited high resistance to β -lactam and quinolone antibiotics (Diwan et al., 2012). In another study conducted in Rio de Janeiro, Brazil, many Enterobacteriaceae isolated from hospital wastewater, showed 48% and 21% resistance to cefotaxime and ciprofloxacin, respectively (Chagas et al., 2011). Similar results with respect to quinolone and cephalosporin-resistant *E. coli* in hospital wastewater were reported from Vietnam (Duong et al., 2008), China (Han et al., 2012), Denmark (Jakobsen et al., 2008) and Poland (Korzeniewska and Harnisz, 2013). 89% of the β -lactamase producing Enterobacteriaceae from hospital effluents in Singapore had multidrug-resistance phenotype including quinolones, trimethoprim, and aminoglycosides (Haller et al., 2018). Resistance to aminoglycosides was found in only 15 H2 isolates in the present study, and this disparity in resistance levels could be due to differences in the geographical area of study and also on the pattern of prescribing antibiotics in local society (Mathai et al., 2008; Sahoo et al., 2012).

Among aquaculture farms, the occurrence of quinolone and cephalosporin resistant isolates was highest in F3, the farm which was very closely located (800 m) to the hospital discharge site followed by F1 and F2. All the isolates from F4 and F5, the farms which were located farthest from the hospital location, were found to be sensitive to most of the antibiotics tested. It must be pointed out that there was no antibiotic use in the aquaculture farms selected for the study during the culture activities. The location of the aquaculture farm appeared to be a critical factor. As the distance between the aquaculture farm and point of discharge of hospital effluent increased, there was a significant decrease in the number of MDR isolates. This may be attributed to the increased dilution under flowing water conditions.

In this study, of the 122 PMQR *E. coli* isolates, 110 isolates had MIC of ciprofloxacin ranging from 8 to 64 $\mu\text{g/mL}$. Studies have shown that mutations in the quinolone-resistance determining regions (QRDRs) induce high level MICs while PMQR genes induce low level MICs (Robicsek et al., 2006c; Strahilevitz et al., 2009). The type of selection pressure that may activate the PMQR genes and enhance or facilitate their horizontal transfer has been a subject of study. Vredenburg et al. (2013) observed PMQR in bacteria isolated from ciprofloxacin-supplemented culture medium. Their observations indicated that

PMQR can confer a selective advantage that is not necessarily linked to an increased MIC value. Indeed, it has been suggested that PMQR can contribute to enhancing the prevalence of resistant mutants in a population by conferring a low level of resistance (Jacoby, 2005). Accordingly, PMQR may then serve as facilitators for the acquisition of mutations, which may be lost at a later stage when they are no longer needed. In addition, the genes were also found in both quinolone-resistant and susceptible isolates, suggesting that their function is to facilitate and enhance the emergence of resistance (Briales et al., 2012; Park et al., 2006; Zurfluh et al., 2014). The presence of *qnr* or *aac* (6')-*Ib-cr* proteins is known to promote the selection of resistance mutations in the presence of quinolone concentrations that would otherwise be lethal (Kim et al., 2009a). Similar to the results obtained in the present study, Rodriguez-Martinez et al. (2011) also reported MIC of ciprofloxacin between 8 and 64 $\mu\text{g/mL}$ in isolates with PMQR genes.

Earlier studies have indicated that PMQR was uncommon in the environment. However, in this study, we found ciprofloxacin-resistant *E. coli* isolates with *qnr* genes on plasmids, indicating that plasmid-mediated quinolone resistance in this setting is not unusual. There are very few reports of quinolone resistance genetic determinants in *E. coli* isolated from hospital wastewater or commensal or clinical *E. coli* isolates from India (Diwan et al., 2012; Magesh et al., 2011; Pathak et al., 2013). In the present study, among 122 isolates of *E. coli* resistant to quinolones from the direct hospital effluents and farms, *qnrB* was most prevalent (33%) followed by *qnrS* (20.4%) and *qnrA* (1.6%). This result supports the findings of other researchers suggests that isolates obtained from different clinical samples, *qnrB* was more prevalent than other *qnr* genes (Cattoir et al., 2007; Robicsek et al., 2006c; Strahilevitz et al., 2009; Wu et al., 2008). Similarly, a study by Kim et al. (2009b) also found that *qnrB* was the most prevalent PMQR gene observed among clinical Enterobacteriaceae isolates of tertiary care hospital in the Republic of Korea. A study by Kindle et al. (2019) reported the presence of *E. coli* with quinolone resistance in environmental samples and showed that *qnrB* is the most frequent PMQR gene. They opined that the presence of the *qnr* genes in *E. coli* indicates that the selection may occur without exposure to inhibitory concentrations of fluoroquinolone. *E. coli* isolates with *qnrB* and *qnrS* resistance genes was also detected from three major hospitals in Tehran, Iran by Abbasi and Ranjbar (2018). An important finding in our study was the detection of *qnrA* and *aac* (6')-*Ib-cr* which can confer increased resistance to fluoroquinolones in addition to aminoglycosides. Tarchouna et al. (2015) detected plasmid-mediated quinolone resistance in clinical isolates of *E. coli* in a Tunisian hospital and reported a frequency of 32% of *qnr* genes, 12.5% for *qnrB*, 5.3% for *qnrA* and 3.5% for *qnrS*. *aac* (6')-*Ib-cr*, a variant of aminoglycoside acetyl transferase capable of modifying and decreasing ciprofloxacin activity, was commonly reported and observed to be distributed in combination with *qnr* genes (Magesh et al., 2011). The presence of *aac* (6')-*Ib-cr* plasmid-mediated quinolone resistance determinant was identified in an Italian hospital exclusively in *E. coli* isolates and the gene was located on a plasmid which presented additional ESBL genes (Frasson et al., 2011). A study by Bartley et al. (2019) reported the presence of PMQR genes including *qnrS* and *aac* (6')-*Ib-cr* in Enterobacteriaceae samples from an urban lake in Brazil and this study pointed out poor sanitation may be a component of the resistome. A study by Berglund (2015) points out that poor sanitation may promote the emergence, dissemination and persistence of AMR even where selective pressure from human antibiotic consumption is not intense.

Another interesting finding in this research has been the detection of *OqxAB*, a plasmid-borne efflux pump gene, in six *E. coli* isolates from H2 sampling site. Their phenotypic activity was confirmed by high fluorescence in the EtBr plate with a concentration of 2.5 mg/L in the agar cart-wheel method. *OqxAB* is a member of the resistance-nodulation-cell division (RND) family of multidrug efflux pumps. A novel plasmid-coded multi-drug efflux pump *OqxAB* was first reported in *E. coli* of swine manure in Denmark in 2004 (Hansen et al., 2004). Over the last decades the prevalence of *OqxAB* in Enterobacteriaceae has been

increasingly reported (Perez et al., 2013). *OqxAB* was encoded with *OqxA* and *OqxB* genes located in a 52-kb plasmid designated as pOLA52. Its overexpression confers resistance to olaquinox, trimethoprim, chloramphenicol and decreases bacterial susceptibility to fluoroquinolones, increasing the MICs with these agents (Hansen et al., 2007; Norman et al., 2008). Sorensen et al. (2003) isolated an *E. coli* with *OqxAB* from pork manure in a farm using olaquinox as an additive for feed. The prevalence of *OqxAB* in human-origin *E. coli* was first identified in 2009 (Kim et al., 2009c). *OqxAB* was reported to be present in 7% of *E. coli* and in all clinically isolated *Klebsiella pneumoniae* strains in a study reported from China (Yuan et al., 2012). *OqxAB* genes were found in 20% of human (feces and urine), animal (heart, liver, spleen, blood, or feces samples) and environmental (different farm soil, sewage, drinking water and pond water) isolates of *E. coli* collected in China between 1993 and 2010 (Chen et al., 2012). A study by Chandran et al. (2014) detected 190 *E. coli* isolates from rural and urban hospital wastewaters in Ujjain, India. A total of 44% were quinolone resistant, among them two isolates were positive for *OqxAB*. The *OqxAB* multidrug efflux pump is also known to lead to a reduced susceptibility to detergents and disinfectants, including benzalkonium chloride, triclosane, especially SDS (Hansen et al., 2007). Being plasmid borne, the overexpression of this gene can confer multi-drug resistance to different bacteria (Li et al., 2019) and therefore, poses a risk to public health as it could facilitate the development and transfer of AMR. The other quinolone genes *qnrC*, *qnrD* and quinolone resistance-determining regions (QRDRs) genes were not detected in any of the quinolone resistant *E. coli* isolates in the present study. This may be because of other mechanisms such as altering the expression of outer membrane porin proteins *ompF*, *ompC* and other efflux pump encoding genes in *E. coli* such as *AcrAB-TolC*, that were not analyzed in this study, could have also contributed to the final phenotype. Cephalosporin resistance was primarily due to ESBL production and *E. coli* isolates from H1, H2 and F3 were positive for *bla_{CTX-M}* and *bla_{SHV}* genes. Most of the *bla_{CTX-M}* positive *E. coli* isolates from direct hospital effluents co-harboured *bla_{SHV}* gene also (data not shown).

An unusual finding in this study was the detection of the *qacE Δ1* gene in a single *E. coli* isolate from the H2 sampling site. Gram-negative bacteria have identified efflux pumps, *qacE*, *qacE Δ1*, *qacF*, *qacG*, *qacH/I* and *SugE(p)* which contributed resistance to quarternary ammonium compounds and they are generally located on mobile genetic elements such as integrons and plasmids (Zou et al., 2014). *qac* genes are closely associated with Class 1 integrons and *qacE* is located in the Class 1 integrons 3'-conserved segment (CS). *qacE Δ1*, a deletion derivative of *qacE* confers higher resistance to Benzalkonium Chloride, a commonly used disinfectant in farms and food processing environment (Kazama et al., 1998). Massive use of QACs in the farm environment may lead to acquired resistance of QACs in strains of *E. coli* (Sidhu et al., 2002). Many resistant genes of QACs are commonly associated with multidrug resistant pathotypes, especially *qacC/D*, *qacA/B* and *qacE* (Zhang et al., 2015). The *qacE Δ1* gene is widespread in enteric bacterial pathogens with resistant determinants of sulphonamide (Kucken et al., 2000).

The plasmid mediated quinolone resistant gene, *qnrS* was successfully transferred to *E. coli* J53. Transconjugant *E. coli* J53 with ciprofloxacin resistance clearly shows that resistance can be transferred easily by plasmids between *E. coli*. Therefore, exposure to antibiotics for a longer duration of time may be an ideal condition for selection and normal aquatic bacteria can acquire resistance through conjugation and other transfer mechanisms. The detection of resistance genes in *E. coli* isolates in direct hospital effluents and aquaculture farms indicate the possibility of transmission and dissemination to other bacteria through plasmid mediated horizontal transfer under selection pressure of antibiotics. Studies revealed that abundance of nutrients in wastewater, high microbial density and its diversity, biofilm variety and activated sludge provides an ideal condition for horizontal gene transfer (Kelly et al., 2009).

Having observed consistently high resistance to quinolones in the *E. coli* isolates from direct hospital effluents, a cursory analysis of the water and sediment samples from these locations was carried out for the presence of quinolone residues in an UPLC-MS/MS. Among the antibiotics analyzed, ciprofloxacin (CIP) residue was detected in the sediment sample from H2 hospital discharge site at a level of 15 ppb. The minimum selective concentration (MSC) for CIP, a second-generation fluoroquinolone, was reported to be 0.1 ppb (Gullberg et al., 2011). According to the guidelines of the Ministry of Environment, Forest and Climate Change, Government of India (2020), Environment (Protection) Amendment Rules, 2019, the limiting value for ciprofloxacin residues in the treated effluent should be 0.02 ppb. In our study, CIP level detected in one hospital discharge point was 15 ppb; this relatively high concentration is a matter of serious concern. In a detailed probabilistic hazard assessment of the results of various studies on CIP levels from different parts of the world, Kelly and Brooks (2018) found ciprofloxacin in both raw and treated hospital effluent in all geographic regions. In hospital wastewater, elevated levels of ciprofloxacin were identified compared with municipal wastewater and effluent. Most of the detections reported were in Asia, followed by Europe.

Residue detection can be explained in two ways: it is one of the most prescribed antibiotics for urinary tract infections by Gram-negative bacteria in hospitals and is not readily biodegradable in the environment (Weller et al., 2011). Previous studies showed that most of the quinolone class of antibiotics is excreted in un-metabolized form. Non-metabolized antibiotics commonly contain their active groups that have not been properly metabolized or degraded, and this may have a significant impact on the aquatic environment (Giger et al., 2007). Massmann et al. (2008) have shown that antibiotics may persist in aquatic environments for decades. Their low degradability and high potential for binding to sediments promote environmental pollution with residual fluoroquinolones (Marengo et al., 1997) and some fluoroquinolones remain stable for at least 120 days in the absence of solar radiation (Wu et al., 2005). Though in this study only one sediment sample showed the presence of fluoroquinolone, the limit of quantitation (LOQ) of the analysis carried out was 10 ppb. So, it may also be possible that levels of antibiotic residues in the samples from the other locations could have been lower than the detectable levels and not necessarily absent. Also, factors such as dilution of water and other environmental factors like salinity may have a role in persistence.

It is claimed that in aquatic systems antibiotic residues are reduced relative to their therapeutic concentration, and thus can be harmless. Nevertheless, exposure to concentrations below the sub therapeutic level over long periods of time may be an optimal condition for selection and consequent resistance spreading (Kummerer and Henninger, 2003). Antibiotics at low sub-inhibitory concentration may also affect cell functions, alter the genetic expression of virulence factors and promote the transfer of antibiotic resistance (Kummerer, 2009). High residue levels of fluoroquinolones can exert genotoxic effects in the aquatic environment (Diwan et al., 2010). The presence of CIP residue in the sediment sample, in the present study, from one of the location at levels much higher than the acceptable range, indicates the frequent use in hospitals and increased discharge of fluoroquinolones to the water bodies. This is probably one of the main driving factors for the high quinolone resistance found in the *E. coli* isolates from the region.

A study by Diwan et al. (2010) highlighted the fact that in low- and middle-income countries, antibiotics reach the aquatic environment mainly through hospital effluents, which do not undergo any treatment process. Even in the high income-countries, antibiotics have been detected in hospital effluents (Kummerer, 2009). Ciprofloxacin was the most frequently detected antibiotic among fluoroquinolones in hospital effluents (Lundborg and Tamhankar, 2017; Mahmood et al., 2018). They were also detected in different environmental samples such as well, surface water, tap water, wastewater, sediments, pig manure, soil and vegetables in eastern China (Hanna et al., 2018). All this demonstrates that there is a need to develop effluent standards specific for healthcare

facilities and current WWTPs should be complemented by innovative treatment strategies which take into account Mobile Genetic Elements (MGEs), ARGs and antibiotic residues, as suggested by Barancheshme and Munir (2018).

5. Conclusion

The present study displays the potential for the spread of quinolone resistant *E. coli* in aquatic environment, where antibiotic pressure is very high. The existence of quinolone resistance in *E. coli* from the direct hospital effluents and adjoining aquaculture farms is disconcerting and points to the widespread use in hospital settings. Exposure to lower antibiotic levels increases the probability of resistance selection. They may therefore be the source of a significant driving force for quinolone resistance selection and all the cited hospitals are located in crowded cities. The wastewater from these systems can readily contaminate the environment leading to the spread of multidrug-resistant bacteria. The comparatively elevated levels of quinolone-resistant bacterial pathogens observed in sediment samples near the direct hospital effluent discharge site support the conviction that hospital wastewater is a significant contributor to antibiotic loads in aquatic environment. The study emphasizes the need for constant monitoring of hospital wastewater for the presence of antibiotic residues, antibiotic-resistant bacteria and the immediate need for effective wastewater treatment plants in hospitals as part of antibiotic stewardship schemes. This study also points to a deeper systemic problem for a state like Kerala which has a natural abundance of water bodies, where hospitals are key hotspots for dissemination of AMR bacteria. The warm temperatures and high relative humidity aid in the survival of such hospital superbugs and emergence of new ones in the environment, and is an occupational and food hazard not just for fishermen and farmers, but eventually the community at large.

CRediT authorship contribution statement

Sneha Kalasseril Girijan: performing the experiments, Writing - original draft, Writing - review and editing. **Robin Paul:** Writing - review and editing. **Rejish Kumar:** helped in Editing. **Devika Pillai:** Supervision, conception and planning of the study, critical review and editing of the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors wish to thank the authorities of the Kerala University of Fisheries and Ocean Studies for providing funds and facilities to carry out the work. We also thank the Export Inspection Agency, Kerala, India for the analysis of samples for antibiotic residues. This research did not receive any specific grant from funding agencies in the public, commercial or not for profit sectors.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2020.141538>.

References

Abbasi, H., Ranjbar, R., 2018. The prevalence of quinolone resistance genes of A, B, S in *Escherichia coli* strains isolated from three major hospitals in Tehran, Iran. *Cent. European. J. Urol.* 71, 129–133. <https://doi.org/10.5173/cej.2018.1539>.

Adefisoye, M.A., Okoh, A.J., 2016. Identification and antimicrobial resistance prevalence of pathogenic *Escherichia coli* strains from treated wastewater effluents in Eastern Cape, South Africa. *Microbiologyopen* 5, 143–151. <https://doi.org/10.1002/mbo3.319>.

Azuma, T., Otomo, K., Kunitou, M., Shimizu, M., Hosomaru, K., Mikata, S., Ishida, M., Hisamatsu, K., Yunoki, A., Mino, Y., Hayashi, T., 2019. Environmental fate of pharmaceutical compounds and antimicrobial-resistant bacteria in hospital effluents, and contributions to pollutant loads in the surface waters in Japan. *Sci. Total Environ.* 657, 476–484. <https://doi.org/10.1016/j.scitotenv.2018.11.433>.

Barancheshme, F., Munir, M., 2018. Strategies to combat antibiotic resistance in the wastewater treatment plants. *Front. Microbiol.* 8, 1–12. <https://doi.org/10.3389/fmicb.2017.02603>.

Bartley, P.S., Domitrovic, T.N., Moretto, V.T., Santos, C.S., Ponce-Terashima, R., Reis, M.G., Barbosa, L.M., Blanton, R.E., Bonomo, R.A., Perez, F., 2019. Antibiotic resistance in Enterobacteriaceae from surface waters in urban Brazil highlights risks of poor sanitation. *Am. J. Trop. Med. Hyg.* 100, 1369–1377. <https://doi.org/10.4269/ajtmh.18-0726>.

Bauer, A.W., Kirby, W.M., Sherris, J.C., Turck, M., 1966. *Antibiotic susceptibility testing by standardized single disc method*. *Am. J. Clin. Pathol.* 44, 493–496 (PMID: 5325707).

Bell, B.G., Schellevis, F., Stobberingh, E., Goossens, H., Pringle, M., 2014. A systematic review and meta-analysis of the effects of antibiotic consumption on antibiotic resistance. *BMC Infect. Dis.* 9, 1–25. <https://doi.org/10.1186/1471-2334-14-13>.

Berglund, B., 2015. Environmental dissemination of antibiotic resistance genes and correlation to anthropogenic contamination with antibiotics. *Infect. Ecol. Epidemiol.* 5, 28564. <https://doi.org/10.3402/iee.v5.28564>.

Briales, A., Rodríguez-Martínez, J.M., Velasco, C., de-Alba, P.D., Rodríguez-Bano, J., Martínez-Martínez, L., Pascual, A., 2012. Prevalence of plasmid-mediated quinolone resistance determinants *Qnr* and *aac(6′)-Ib-cr* in *Escherichia coli* and *Klebsiella pneumoniae* producing extended-spectrum β -lactamases in Spain. *Int. J. Antimicrob. Agents* 39, 431–434. <https://doi.org/10.1016/j.ijantimicag.2011.12.009>.

Caekenbergh, D.L.V., Pattyn, S.R., 1984. In vitro activity of ciprofloxacin compared with those of other new fluorinated piperazinyl-substituted quinoline derivatives. *Antimicrob. Agents Chemother.* 25, 518–521. <https://doi.org/10.1128/aac.25.4.518>.

Castillo, F.Y.R., Gonzalez, F.J.A., Garneau, P., Diaz, F.M., Barrera, A.L.G., 2013. Presence of multi-drug resistant pathogenic *Escherichia coli* in the San Pedro River located in the State of Aguascalientes, Mexico. *Front. Microbiol.* 4, 1–16. <https://doi.org/10.3389/fmicb.2013.00147>.

Cattoir, V., Poirel, L., Rotimi, V., Soussy, C.J., Nordmann, P., 2007. Multiplex PCR for detection of plasmid-mediated quinolone-resistance *Qnr* genes in ESBL-producing Enterobacterial isolates. *J. Antimicrob. Chemother.* 60, 394–397. <https://doi.org/10.1093/jac/dkm204>.

Cavaco, L.M., Hasman, H., Xia, S., Aarestrup, F.M., 2009. *qnrD*, a novel gene conferring transferable quinolone resistance in *Salmonella enterica* serovar Kentucky and *Bovismorbificans* strains of human origin. *Antimicrob. Agents Chemother.* 53, 603–608. <https://doi.org/10.1128/AAC.00997-08>.

CDDEP, 2015. *ResistanceMap* Washington DC: Center for Disease Dynamics, Economics & Policy. (August 20, 2015). <http://www.resistancemap.org>.

Chagas, T.P.G., Seki, L.M., Cury, J.C., Oliveira, J.A.L., Davila, A.M.R., Silva, D.M., Asensi, M.D., 2011. Multiresistance beta-lactamase-encoding genes and bacterial diversity in hospital wastewater in Rio de Janeiro, Brazil. *J. Appl. Microbiol.* 111, 572–581. <https://doi.org/10.1111/j.1365-2672.2011.05072.x>.

Chandran, S.P., Diwan, V., Tamhankar, A.J., Joseph, B.V., Rosales-Klintz, S., Mundayoor, S., Lundborg, C.S., Macaden, R., 2014. Detection of carbapenem resistance genes and cephalosporin, and quinolone resistance genes along with *oqxAB* gene in *Escherichia coli* in hospital wastewater: a matter of concern. *J. Appl. Microbiol.* 117, 984–995. <https://doi.org/10.1111/jam.12591>.

Chen, X., Zhang, W., Pan, W., Yin, J., Pan, Z., Gao, S., Jiao, X., 2012. Prevalence of *qnr*, *aac(6′)-Ib-cr*, *qepA*, and *oqxAB* in *Escherichia coli* isolates from humans, animals, and the environment. *Antimicrob. Agents Chemother.* 56, 3423–3427. <https://doi.org/10.1128/AAC.06191-11>.

CLSI, Clinical and Laboratory Standards Institute, 2013. *Performance Standards for Antimicrobial Susceptibility Testing; Twenty-Third Informational Supplement*. Document M100-S23. Wayne, PA.

Diwan, V., Tamhankar, A.J., Khandal, R.K., Sen, S., Aggarwal, M., Marothi, Y., Iyer, R.V., Sundblad-Tonderski, K., Stalsby-Lundborg, C., 2010. Antibiotics and antibiotic-resistant bacteria in waters associated with a hospital in Ujjain, India. *BMC Public Health* 10, 414. <https://doi.org/10.1186/1471-2458-10-414>.

Diwan, V., Chandran, S.P., Tamhankar, A.J., Lundborg, C.S., Macaden, R., 2012. Identification of extended-spectrum β -lactamase and quinolone resistance genes in *Escherichia coli* isolated from hospital wastewater from central India. *J. Antimicrob. Chemother.* 67, 857–859. <https://doi.org/10.1093/jac/dkr564>.

Duong, H.A., Pham, N.H., Nguyen, H.T., Hoang, T.T., Pham, H.V., Pham, V.C., Berg, M., Giger, W., Alder, A.C., 2008. Occurrence, fate and antibiotic resistance of fluoroquinolone antibacterials in hospital wastewaters in Hanoi, Vietnam. *Chemosphere* 72, 968–973. <https://doi.org/10.1016/j.chemosphere.2008.03.009>.

Environment (Protection) Amendment Rules, 2019. Ministry of Environment, Forest and Climate Change Notification (New Delhi, January 2020).

Farooqui, H.H., Mehta, A., Selvaraj, S., 2019. Outpatient antibiotic prescription rate and pattern in the private sector in India: evidence from medical audit data. *PLoS One* 14, e0224848. <https://doi.org/10.1371/journal.pone.0224848>.

Frasconi, I., Cavallaro, A., Bergo, C., Richter, S.N., Palu, G., 2011. Prevalence of *aac(6′)-Ib-cr* plasmid-mediated and chromosome-encoded fluoroquinolone resistance in Enterobacteriaceae in Italy. *Gut. Pathog.* 3, 12. <https://doi.org/10.1186/1757-4749-3-12>.

Galvin, S., Boyle, F., Hickey, P., Vellinga, A., Morris, D., Cormican, M., 2010. Enumeration and characterization of antimicrobial-resistant *Escherichia coli* bacteria in effluent from municipal, hospital, and secondary treatment facility sources. *Appl. Environ. Microbiol.* 76, 4772–4779. <https://doi.org/10.1128/AEM.02898-09>.

- Gandra, S., Kotwani, A., 2019. Need to improve availability of access group antibiotics and reduce the use of watch group antibiotics in India for optimum use of antibiotics to contain antimicrobial resistance. *J. Pharm. Policy. Pract.* 12, 20. <https://doi.org/10.1186/s40545-019-0182-1>.
- Giger, W., Alder, A.C., Gobel, A., Golet, E., Huber, M., Joss, A., Keller, E., Kohler, H.P., McArdell, C.S., et al., 2007. Behaviour and Removal of Antibiotics in Wastewater Treatment. Presented at 1st EMCO (Emerging Contaminants) Workshop 'Analysis and Removal of Contaminants From Wastewaters for the Implementation of the Water Framework Directive (WFD)', European Union Sixth Framework Programme October 21–25, 2005. Dubrovnik, Croatia.
- Gniadkowski, M., 2001. Evolution and epidemiology of extended-spectrum β -lactamases (ESBLs) and ESBL-producing microorganisms. *Clin. Microbiol. Infect.* 7, 597–608. <https://doi.org/10.1046/j.1198-743x.2001.00330.x>.
- Gullberg, E., Cao, S., Berg, O.G., Ilbäck, C., Sandegren, L., Hughes, D., Andersson, D.I., 2011. Selection of resistant bacteria at very low antibiotic concentrations. *PLoS Pathog.* 7, e1002158. <https://doi.org/10.1371/journal.ppat.1002158>.
- Haberecht, H.B., Nealon, N.J., Gilliland, J.R., Holder, A.V., Runyan, C., Oppel, R.C., Ibrahim, H.M., Mueller, L., Schrupp, F., Vilchez, S., Antony, L., Scaria, J., Ryan, E.P., 2019. Antimicrobial resistant *Escherichia coli* from environmental waters in Northern Colorado. *J. Environ. Public Health* 2019, 1–3. <https://doi.org/10.1155/2019/3862949>.
- Haller, L., Chen, H., Ng, C., Le, T.H., Koh, T.H., Barkham, T., 2018. Occurrence and characteristics of extended-spectrum β -lactamase- and carbapenemase-producing bacteria from hospital effluents in Singapore. *Sci. Total Environ.* 615, 1119–1125. <https://doi.org/10.1016/j.scitotenv.2017.09.217>.
- Han, N., Sheng, D., Xu, H., 2012. Role of *Escherichia coli* strain subgroups, integrons, and integron-associated gene cassettes in dissemination of antimicrobial resistance in aquatic environments of Jinan, China. *Water Sci. Technol.* 66, 2385–2392. <https://doi.org/10.2166/wst.2012.473>.
- Hanna, N., Sun, P., Sun, Q., Li, X., Yang, X., Ji, X., Zou, H., Ottoson, J., Nilsson, L.E., Berglund, B., Dyar, O.J., Tamhankar, A.J., Lundborg, C.S., 2018. Presence of antibiotic residues in various environmental compartments of Shandong province in eastern China: its potential for resistance development and ecological and human risk. *Environ. Int.* 114, 131–142. <https://doi.org/10.1016/j.envint.2018.02.003>.
- Hansen, L.H., Johannessen, E., Burmolle, M., Sorensen, A.H., Sorensen, S.J., 2004. Plasmid-encoded multidrug-efflux pump conferring resistance to olaquinoxon on *Escherichia coli*. *Antimicrob. Agent. Chemother.* 48, 3332–3337. <https://doi.org/10.1128/AAC.48.9.3332-3337.2004>.
- Hansen, L.H., Jensen, L.B., Sorensen, H.I., Sorensen, S.J., 2007. Substrate specificity of the OqxAB multidrug-resistance pump in *Escherichia coli* and selected enteric bacteria. *J. Antimicrob. Chemother.* 60, 145–147. <https://doi.org/10.1093/jac/dkm167>.
- Hata, M., Suzuki, M., Matsumoto, M., Takahashi, M., Sato, K., Ibe, S., Sakae, K., 2005. Cloning of a novel gene for quinolone resistance from a transferable plasmid in *Shigella flexneri* 2b. *Antimicrob. Agents Chemother.* 49, 801–803. <https://doi.org/10.1128/AAC.49.2.801-803.2005>.
- Hirsch, R., Ternes, T., Haberer, K., Kratz, K.L., 1999. Occurrence of antibiotics in the aquatic environment. *Sci. Total Environ.* 225, 109–118. [https://doi.org/10.1016/S0048-9697\(98\)00337-4](https://doi.org/10.1016/S0048-9697(98)00337-4).
- Holloway, K., Mathai, E., Sorensen, T., Gray, T., 2009. Community-based Surveillance of Antimicrobial Use and Resistance in Resources-constrained Settings: Report on Five Pilot Projects. World Health Organization, Geneva.
- Holt, J.G., Krieg, N.R., Sneath, P.H.A., Staley, J.T., Williams, S.T., 1994. *Bergey's Manual of Determinative Bacteriology*. Ninth edition. (Maryland, USA).
- Hooper, D.C., 2001. Mechanisms of action of antimicrobials: focus on fluoroquinolones. *Clin. Infect. Dis.* 32, S9–S15. <https://doi.org/10.1086/319370>.
- Jacoby, G.A., 2005. Mechanism of resistance to quinolones. *Clin. Infect. Dis.* 41, S120–S126. <https://doi.org/10.1086/428052>.
- Jakobsen, L., Sandvang, D., Hansen, L.H., Bagger-Skjot, L., Westh, H., Jorgensen, C., Hansen, D.S., Pedersen, B.M., Monnet, D.L., Primodt-Moller, N., Sorensen, S.J., Hammerum, A.M., 2008. Characterization, dissemination and persistence of gentamicin resistant *Escherichia coli* from a Danish university hospital to the wastewater environment. *Environ. Int.* 34, 108–115. <https://doi.org/10.1016/j.envint.2007.07.011>.
- Jang, J., Suh, Y.S., Di, D.Y.W., Unno, T., Sadowsky, M.J., Hur, H.G., 2013. Pathogenic *Escherichia coli* strains producing extended-spectrum β -lactamases in the Yeongsan River basin of South Korea. *Environ. Sci. Technol.* 47, 1128–1136. <https://doi.org/10.1021/es303577u>.
- Jia, W., Wang, J., Xu, H., Li, G., 2015. Resistance of *Stenotrophomonas maltophilia* to fluoroquinolones: prevalence in a university hospital and possible mechanisms. *Int. J. Environ. Res. Public Health* 12, 5177–5195. <https://doi.org/10.3390/ijerph120505177>.
- Jiang, X., Ellabaan, M.M.H., Charusanti, P., Munck, C., Blin, K., Tong, Y., Weber, T., Sommer, M.O.A., Lee, S.Y., 2017a. Dissemination of antibiotic resistance genes from antibiotic producers to pathogens. *Nat. Commun.* 8, 1–7. <https://doi.org/10.1038/ncomms15784>.
- Jiang, X., Yu, T., Liu, L., Li, Y., Zhang, K., Wang, H., Shi, L., 2017b. Examination of quaternary ammonium compound resistance in *Proteus mirabilis* isolated from cooked meat products in China. *Front. Microbiol.* 8, 2417. <https://doi.org/10.3389/fmicb.2017.02417>.
- Kazama, H., Hamashima, H., Sasatsu, M., Arai, T., 1998. Distribution of the antiseptic-resistance gene qacE and qacE11 in Gram-negative bacteria. *FEMS Microbiol. Lett.* 159, 173–178. <https://doi.org/10.1111/j.1574-6968.1998.tb12857.x>.
- Kelly, K.R., Brooks, B.W., 2018. Global aquatic hazard assessment of ciprofloxacin: exceedances of antibiotic resistance development and ecotoxicological thresholds. *Prog. Mol. Biol. Transl. Sci.* 159, 59–77. <https://doi.org/10.1016/bs.pmbts.2018.07.004>.
- Kelly, B.G., Vespermann, A., Bolton, D.J., 2009. Gene transfer events and their occurrence in selected environments. *Food Chem. Toxicol.* 47, 978–983. <https://doi.org/10.1016/j.fct.2008.06.012>.
- Kim, E.S., Jeong, J.Y., Jun, J.B., Choi, S.H., Lee, S.O., Kim, M.N., Woo, J.H., Kim, Y.S., 2009a. Prevalence of aac(6′)-Ib-cr encoding a ciprofloxacin-modifying enzyme among Enterobacteriaceae blood isolates in Korea. *Antimicrob. Agents Chemother.* 53, 2643–2645. <https://doi.org/10.1128/AAC.01534-08>.
- Kim, H.B., Park, C.H., Kim, C.J., Kim, E.C., Jacoby, G.A., Hooper, D.C., 2009b. Prevalence of plasmid-mediated quinolone resistance determinants over a 9-year period. *Antimicrob. Agents Chemother.* 53, 639–645. <https://doi.org/10.1128/AAC.01051-08>.
- Kim, H.B., Wang, M., Park, C.H., Kim, E.C., Jacoby, G.A., Hooper, D.C., 2009c. oqxAB encoding a multidrug efflux pump in human clinical isolates of Enterobacteriaceae. *Antimicrob. Agents Chemother.* 53, 3582–3584. <https://doi.org/10.1128/AAC.01574-08>.
- Kindle, P., Zurfluh, K., Nuesch-Inderbinnen, M., von Ah, S., Sidler, X., Stephan, R., Kummerlen, D., 2019. Phenotypic and genotypic characteristics of *Escherichia coli* with non-susceptibility to quinolones isolated from environmental samples on pig farms. *Porc. Health. Manag.* 28 (5), 9. <https://doi.org/10.1186/s40813-019-0116-y>.
- Klein, E.Y., Van Boeckel, T.P., Martinez, E.M., Pant, S., Gandra, S., Levin, S.A., Goossens, H., Laxminarayan, R., 2018. Global increase and geographic convergence in antibiotic consumption between 2000 and 2015. *PNAS* 115. <https://doi.org/10.1073/pnas.1717295115> E3463–E70.
- Korzeniewska, E., Harnisz, M., 2013. Beta-lactamase producing Enterobacteriaceae in hospital effluents. *J. Environ. Manag.* 123, 1–7. <https://doi.org/10.1016/j.jenvman.2013.03.024>.
- Kotwani, A., Holloway, A., 2011. Trends in antibiotic use among outpatients in New Delhi, India. *BMC Infect. Dis.* 11, 1–9 (doi:1471-2334/11/99).
- Kotwani, A., Holloway, K., 2014. Antibiotic prescribing practice for acute, uncomplicated respiratory tract infections in primary care settings in New Delhi, India. *Tropical Med. Int. Health* 19, 761–768. <https://doi.org/10.1111/tmi.12327>.
- Kourilaba, G., Kourkouni, E., Spyridis, N., Gerber, J.S., Kopsidas, J., Mougkou, K., Lourida, A., Zautis, T.E., 2015. Antibiotics prescribing and expenditures in outpatient paediatrics in Greece, 2010–2013. *J. Antimicrob. Chemother.* 70, 2405–2408. <https://doi.org/10.1093/jac/dkv091>.
- Kucken, D., Feucht, H., Kaulfers, P., 2000. Association of qacE and qacE Δ 1 with multiple resistance to antibiotics and antiseptics in clinical isolates of Gram-negative bacteria. *FEMS Microbiol. Lett.* 183, 95–98. <https://doi.org/10.1111/j.1574-6968.2000.tb08939.x>.
- Kummerer, K., 2009. Antibiotics in the aquatic environment – a review – part II. *Chemosphere* 75, 435–441. <https://doi.org/10.1016/j.chemosphere.2008.12.006>.
- Kummerer, K., Henninger, A., 2003. Promoting resistance by the emission of antibiotics from hospitals and households into effluent. *Clin. Microbiol. Infect.* 9, 1203–1214. <https://doi.org/10.1111/j.1469-0691.2003.00739.x>.
- Lamba, M., Graham, D.W., Ahammad, S.Z., 2017. Hospital wastewater releases of carbapenem-resistance pathogens and genes in urban India. *Environ. Sci. Technol.* 51, 13906–13912. <https://doi.org/10.1021/acs.est.7b03380>.
- Lavilla, S., Gonzalez-Lopez, J.J., Sabate, M., Garcia-Fernandez, A., Larrosa, M.N., Bartolome, R.M., Carattoli, A., Prats, G., 2008. Prevalence of qnr genes among extended-spectrum beta-lactamase producing enterobacterial isolates in Barcelona, Spain. *J. Antimicrob. Chemother.* 61, 291–295. <https://doi.org/10.1093/jac/dkm448>.
- Ledoussal, B., Almstead, J.K., Flaim, S.M., Gallagher, C.P., Gray, J.L., Hu, X.E., Kim, N.K., McKeever, H.D., Miley, C.J., Twinem, T.L., Zheng, S.X., 1999. *Abstr. 39th Intersci. Conf. Antimicrob. Agents Chemother. abstr. F544, 303.*
- Li, J., Zhang, H., Ning, J., Sajid, A., Cheng, G., Yuan, Z., Hao, H., 2019. The nature and epidemiology of OqxAB, a multidrug efflux pump. *Antimicrob. Resist. Infect. Control.* 8, 44. <https://doi.org/10.1186/s13756-019-0489-3>.
- Lundborg, C.S., Tamhankar, A.J., 2017. Antibiotic residues in the environment of South East Asia. *Br. Med. J.* 358, j2440. <https://doi.org/10.1136/bmj.j2440>.
- Ma, J., Zeng, Z., Chen, Z., Xu, X., Wang, X., Deng, Y., Lu, D., Huang, L., Zhang, Y., Liu, J., Wang, M., 2009. High prevalence of plasmid-mediated quinolone resistance determinants qnr, aac(6′)-Ib-cr, and qepA among ceftiofur resistant Enterobacteriaceae isolates from companion and food producing animals. *Antimicrob. Agents Chemother.* 53, 519–524. <https://doi.org/10.1128/AAC.00886-08>.
- Magesh, H., Kamatchi, C., Vaidyanathan, R., Sumathi, G., 2011. Identification of plasmid-mediated quinolone resistance genes qnrA1, qnrB1 and aac(6′)-Ib-cr in a multiple drug resistant isolate of *Klebsiella pneumoniae* from Chennai. *Indian J. Med. Microbiol.* 29, 262–268. <https://doi.org/10.4103/0255-0857.83910>.
- Mahmood, A.R., Al-Haidari, H.H., Hassan, F.M., 2018. Detection of antibiotics in drinking water treatment plants in Baghdad City. *Iraq. Adv. Public. Health.* 2019, 1–10. <https://doi.org/10.1155/2019/7851354>.
- Mammeri, H., Loo, M.V.D., Poirel, L., Martinez-Martinez, L., Nordmann, P., 2005. Emergence of plasmid-mediated quinolone resistance in *Escherichia coli* in Europe. *Antimicrob. Agents Chemother.* 49, 71–76. <https://doi.org/10.1128/AAC.49.1.71-76.2005>.
- Marengo, J.R., Kok, R.A., O'Brien, K., Velagaleti, R.R., Stamm, J.M., 1997. Aerobic biodegradation of (14C)-sarafloxacin hydrochloride in soil. *Environ. Toxicol. Chem.* 16, 462–471. <https://doi.org/10.1002/etc.5620160311>.
- Marti, E., Variatza, E., Balcazar, J.L., 2014. The role of aquatic ecosystems as reservoirs of antibiotic resistance. *Trends Microbiol.* 22, 36–41. <https://doi.org/10.1016/j.tim.2013.11.001>.
- Martinez-Martinez, L., Pascual, A., Jacoby, G.A., 1998. Quinolone resistance from a transferable plasmid. *Lancet* 351, 797–799. [https://doi.org/10.1016/S0140-6736\(97\)07322-4](https://doi.org/10.1016/S0140-6736(97)07322-4).
- Martins, M., Viveiros, M., Couto, I., Costa, S.S., Pacheco, T., Fanning, S., Pages, J.M., Amaral, L., 2011. Identification of efflux pump mediated multidrug resistant bacteria by the ethidium bromide agar cartwheel method. *In Vivo* 25, 171–178 (PMID: 21471531).
- Massmann, G., Sultenfub, J., Dunnier, U., Knappe, A., Taute, T., Pekdeger, A., 2008. Investigation of groundwater residence times during bank filtration in Berlin: a multi-tracer approach. *Hydrol. Process.* 22, 788–801. <https://doi.org/10.1002/hyp.6649>.

- Mathai, E., Chandy, S.J., Thomas, K., Antoniswamy, B., 2008. Antimicrobial resistance surveillance among commensal *Escherichia coli* in rural and urban areas in Southern India. *Tropical Med. Int. Health* 13, 41–45. <https://doi.org/10.1111/j.1365-3156.2007.01969.x>.
- Mazari-Hiriart, M., Ponce-de-Leon, S., Lopez-Vidal, Y., Islas-Macias, P., Amieva-Fernandez, R.I., Quinones-Falconi, F., 2008. Microbiological implications of periurban agriculture and water reuse in Mexico City. *PLoS One* 3, e2305. <https://doi.org/10.1371/journal.pone.0002305>.
- Naber, K., Adam, D., 1998. Classification of fluoroquinolones. *Int. J. Antimicrob. Agents* 10, 255–257. [https://doi.org/10.1016/S0924-8579\(98\)00059-4](https://doi.org/10.1016/S0924-8579(98)00059-4).
- Norman, A., Hansen, L.H., She, Q., Sorensen, S.J., 2008. Nucleotide sequence of pOLA52: a conjugative IncX1 plasmid from *Escherichia coli* which enables bio-film formation and multidrug efflux. *Plasmid* 60, 59–74. <https://doi.org/10.1016/j.plasmid.2008.03.003>.
- Novo, A., Manaia, C.M., 2010. Factors influencing antibiotic resistance burden in municipal wastewater treatment plants. *Appl. Microbiol. Biot.* 87, 1157–1166. <https://doi.org/10.1007/s00253-010-2583-6>.
- Park, C.H., Robicsek, A., Jacoby, G.A., Sahm, D., Hooper, D.C., 2006. Prevalence in the United States of aac(6′)-Ib-cr encoding a ciprofloxacin modifying enzyme. *Antimicrob. Agents Chemother.* 50, 3953–3955. <https://doi.org/10.1128/AAC.00915-06>.
- Pathak, A., Chandran, S.P., Mahadik, K., Macaden, R., Lundborg, C.S., 2013. Frequency and factors associated with carriage of multi-drug resistant commensal *Escherichia coli* among women attending antenatal clinics in Central India. *BMC Infect. Dis.* 13, 199. <https://doi.org/10.1186/1471-2334-13-199>.
- Perez, F., Rudin, S.D., Marshall, S.H., Coakley, P., Chen, L., Kreiswirth, B.N., Rather, P.N., Hujer, A.M., Toltzis, P., van Duin, D., Paterson, D.L., Bonomo, R.A., 2013. QoxAB, a quinolone and olaquinox efflux pump, is widely distributed among multidrug-resistant *Klebsiella pneumoniae* isolates of human origin. *Antimicrob. Agents Chemother.* 57, 4602–4603. <https://doi.org/10.1128/AAC.00725-13>.
- Peterson, L.R., 2001. Quinolone molecular structure-activity relationships: what we have learned about improving antimicrobial activity. *Clin. Infect. Dis.* 33, S180–S186. <https://doi.org/10.1086/321846>.
- Pharma, T.D.M., Ziora, Z.M., Blaskovich, M.A.T., 2009. Quinolone antibiotics. *Med. Chem. Commun.* 10, 1719–1739. <https://doi.org/10.1039/c9md00120d>.
- Poirel, L., Leviandier, C., Nordmann, P., 2006. Prevalence and genetic analysis of plasmid-mediated quinolone resistance determinants QnrA and QnrS in Enterobacteriaceae isolates from a French university hospital. *Antimicrob. Agents Chemother.* 50, 3992–3997. <https://doi.org/10.1128/AAC.00597-06>.
- Proia, L., Anzil, A., Borrego, C., Farre, M., Llorca, M., Sanchis, J., Bogaerts, P., Balcazar, J.L., Servais, P., 2018. Occurrence and persistence of carbapenemase genes in hospital and wastewater treatment plants and propagation in the receiving river. *J. Hazard. Mater.* 358, 33–43. <https://doi.org/10.1016/j.jhazmat.2018.06.058>.
- Ranjbar, R., Farahani, O., 2017. The prevalence of plasmid-mediated quinolone resistance genes in *Escherichia coli* isolated from hospital wastewater sources in Tehran, Iran. *Iran. J. Public Health* 46, 1285–1291 (PMCID: PMC5632331).
- Robicsek, A., Jacoby, G.A., Hooper, D.C., 2006a. The worldwide emergence of plasmid-mediated quinolone resistance. *Lancet Infect. Dis.* 6, 629–640. [https://doi.org/10.1016/S1473-3099\(06\)70599-0](https://doi.org/10.1016/S1473-3099(06)70599-0).
- Robicsek, A., Strahilevitz, J., Jacoby, G.A., Macielag, M., 2006b. Fluoroquinolone modifying enzyme: a new adaptation of a common aminoglycoside acetyltransferase. *Nat. Med.* 12, 83–88. <https://doi.org/10.1038/nm1347>.
- Robicsek, A., Strahilevitz, J., Sahm, D.F., Jacoby, G.A., Hooper, D.C., 2006c. Qnr prevalence in ceftazidime-resistant Enterobacteriaceae isolates from the United States. *Antimicrob. Agents Chemother.* 50, 2872–2874. <https://doi.org/10.1128/AAC.01647-05>.
- Rodriguez-Martinez, J.M., Cano, M.E., Velasco, C., Martinez-Martinez, L., Pascual, A., 2011. Plasmid-mediated quinolone resistance: an update. *J. Infect. Chemother.* 17, 149–182. <https://doi.org/10.1007/s10156-010-0120-2>.
- Sahoo, K.C., Tamhankar, A.J., Sahoo, S., Sahu, P.S., Klintz, S.R., Lundborg, C.S., 2012. Geographical variation in antibiotic-resistant *Escherichia coli* isolates from stool, cow-dung and drinking water. *Int. J. Environ. Res. Public Health* 9, 746–759. <https://doi.org/10.3390/ijerph9030746>.
- Saxena, S.K., Rangasamy, R., Krishnan, A.A., Singh, D.P., Uke, S.P., Malekad, P.K., Sengar, A.S., Mohamed, D.P., Gupta, A., 2018. Simultaneous determination of multi-residue and multi-class antibiotics in aquaculture shrimps by UPLC-MS/MS. *Food Chem.* 260, 36–343. <https://doi.org/10.1016/j.foodchem.2018.04.018>.
- Sharma, J., Sharma, M., Ray, P., 2010. Detection of TEM & SHV genes in *Escherichia coli* & *Klebsiella pneumoniae* isolates in a tertiary care hospital from India. *Indian J. Med. Res.* 132, 332–336 PMID: 20847381.
- Shetty, S.S., Deekshit, V.K., Jazeela, K., Vittal, R., Rohit, A., Chakraborty, A., Karunasagar, I., 2019. Plasmid-mediated fluoroquinolone resistance associated with extra intestinal *Escherichia coli* isolates from hospital samples. *Indian J. Med. Res.* 149, 192–198. https://doi.org/10.4103/ijmr.IJMR.2092_17.
- Shibl, A.M., Al-Agamy, M.H., Khubnani, H., Senok, A.C., Tawfik, A.F., Livermore, D.M., 2012. High prevalence of acquired quinolone resistance genes among Enterobacteriaceae from Saudi Arabia with CTX-M-15 β -lactamase. *Diagn. Microbiol. Infect. Dis.* 73, 350–353. <https://doi.org/10.1016/j.diagmicrobio.2012.04.005>.
- Sidhu, M.S., Sorum, H., Holck, A., 2002. Resistance to quaternary ammonium compounds in food-related bacteria. *Microb. Drug Resist.* 8, 393–399. <https://doi.org/10.1089/10766290260469679>.
- Sorensen, A.H., Hansen, L.H., Johannesen, E., Sorensen, S.J., 2003. Conjugative plasmid conferring resistance to olaquinox. *Antimicrob. Agents Chemother.* 47, 798–799. <https://doi.org/10.1128/aac.47.2.798-799.2003>.
- Strahilevitz, J., Jacoby, G.A., Hooper, D.C., Robicsek, A., 2009. Plasmid-mediated quinolone resistance: a multifaceted threat. *Clin. Microbiol. Rev.* 22, 664–689. <https://doi.org/10.1128/CMR.00016-09>.
- Szczepanowski, R., Linke, B., Krahn, I., Gartemann, K.H., Gutzkow, T., Eichler, W., Puhler, A., Schluter, A., 2009. Detection of 140 clinically relevant antibiotic resistance genes in the plasmid metagenome of wastewater treatment plant bacteria showing reduced susceptibility to selected antibiotics. *Microbiology* 155, 2306–2319. <https://doi.org/10.1099/mic.0.028233-0>.
- Tacao, M., Correia, A., Henriques, I., 2012. Resistance to broad-spectrum antibiotics in aquatic systems: anthropogenic activities modulate the dissemination of bla(CTX-M)-like genes. *Appl. Environ. Microbiol.* 78, 4130–4140. <https://doi.org/10.1128/AEM.00359-12>.
- Tarchouna, M., Ferjani, A., Marzouk, M., Guedda, I., Boukadida, J., 2015. Prevalence of plasmid-mediated quinolone resistance determinants among clinical isolates of *Escherichia coli* in a Tunisian Hospital. *Int. J. Curr. Microbiol. App. Sci.* 4, 195–206.
- Van Boeckel, T.P., Gandra, S., Ashok, A., Caudron, Q., Grenfell, B.T., Levin, S.A., Laxminarayan, R., 2014. Global antibiotic consumption 2000 to 2010: an analysis of national pharmaceutical sales data. *Lancet Infect. Dis.* 14, 742–750. [https://doi.org/10.1016/S1473-3099\(14\)70780-7](https://doi.org/10.1016/S1473-3099(14)70780-7).
- Van Boeckel, T.P., Brower, C., Gilbert, M., Grenfell, B.T., Levin, S.A., Robinson, T.P., Reillanta, A., Laxminarayan, R., 2015. Global trends in antimicrobial use in food animals. *PNAS* 112, 5649–5654. <https://doi.org/10.1073/pnas.1503141112>.
- Villegas, M.V., Correa, A., Perez, F., Zuluaga, T., Radice, M., Gutkind, G., Casellas, J.M., Ayala, J., Lolans, K., Quinn, J.P., 2004. Colombian Nosocomial Resistance Study Group. CTX-M-12 β -Lactamase in a *Klebsiella pneumoniae* clinical isolate in Colombia. *Antimicrob. Agents Chemother.* 48, 629–631. <https://doi.org/10.1128/aac.48.2.629-631.2004>.
- Vredenburg, J., Varela, A.R., Hasan, B., Bertilsson, S., Olsen, B., da-Rocha, C.N., Bonnedahl, J., Steedt, J., Costa, P.M.D., Manaia, C.M., 2013. Quinolone resistant *Escherichia coli* isolated from birds of prey in Portugal are genetically distinct from those isolated from water environments and gulls in Portugal, Spain and Sweden. *Environ. Microbiol.* 16, 995–1004. <https://doi.org/10.1111/1462-2920.12231>.
- Wang, A., Yang, Y., Lu, Q., Wang, Y., Chen, Y., Deng, L., Ding, H., Deng, Q., Zhang, H., Wang, C., Liu, L., Xu, X., Wang, L., Shen, X., 2008. Presence of qnr gene in *Escherichia coli* and *Klebsiella pneumoniae* resistant to ciprofloxacin isolated from pediatric patients in China. *BMC Infect. Dis.* 8, 68. <https://doi.org/10.1186/1471-2334-8-68>.
- Wang, M., Guo, Q., Xu, X., Wang, X., Ye, X., Wu, S., Hooper, D.C., Wang, M., 2009. New plasmid-mediated quinolone resistance gene, qnrC, found in a clinical isolate of *Proteus mirabilis*. *Antimicrob. Agents Chemother.* 53, 1892–1897. <https://doi.org/10.1128/AAC.01400-08>.
- Weller, M.F., Roma, M.J., Wentzell, S.L., 2011. Removal of Ciprofloxacin from Water Using Adsorption, UV Photolysis and UV/H₂O₂ Degradation. Worcester Polytechnic Institute, Worcester, MA, USA.
- WHO, 2006. Guidelines for the Safe Use of Wastewater, Excreta and Greywater. World Health Organization, Geneva.
- WHO, 2019. Guidelines for the Ten Threats to Global Health in 2019. World Health Organization, Geneva.
- World Health Organization, 2009. Community-Based Surveillance of Antimicrobial Use and Resistance in Resource-Constrained Settings Report on Five Pilot Projects. World Health Organization, Geneva.
- Wu, Y.B., Wang, Z.S., Liao, X.D., 2005. Study on the excretion of enrofloxacin in chicken and its degradation in chicken feces. *Acta. Vet. Zoot. Sin.* 36, 1069–1074.
- Wu, J.J., Ko, W.C., Tsai, S.H., Yan, J.J., 2007. Prevalence of plasmid-mediated quinolone resistance determinants QnrA, QnrB, and QnrS among clinical isolates of Enterobacteriaceae in a Taiwanese hospital. *Antimicrob. Agents Chemother.* 51, 1223–1227. <https://doi.org/10.1128/AAC.01195-06>.
- Wu, J.J., Ko, W.C., Wu, H.M., Yan, J.J., 2008. Prevalence of Qnr determinants among bloodstream isolates of *Escherichia coli* and *Klebsiella pneumoniae* in a Taiwanese hospital. *J. Antimicrob. Chemother.* 61, 1234–1239. <https://doi.org/10.1093/jac/dkn111>.
- Wu, H., Wang, M., Liu, Y., Wang, X., Wang, Y., Lu, J., Xu, H., 2016. Characterization and antimicrobial resistance in *Klebsiella* species isolated from chicken broilers. *Int. J. Food. Microbiol.* 232, 95–102. <https://doi.org/10.1016/j.ijfoodmicro.2016.06.001>.
- Xia, R., Ren, Y., Xu, H., 2013. Identification of plasmid-mediated quinolone resistance qnr genes in multidrug resistant Gram-negative bacteria from hospital wastewaters and receiving waters in the Jinan area, China. *Microb. Drug Resist.* 19, 446–456. <https://doi.org/10.1089/mdr.2012.0210>.
- Yamane, K., Wachino, J., Suzuki, S., Kimura, K., Shibata, N., Kato, H., Shibayama, K., Konda, T., Arakawa, Y., 2007. New plasmid-mediated fluoroquinolone efflux pump, QepA, found in an *Escherichia coli* clinical isolate. *Antimicrob. Agents Chemother.* 51, 3354–3360. <https://doi.org/10.1128/AAC.00339-07>.
- Yang, H., Chen, H., Yang, Q., Chen, M., Wang, H., 2008. High prevalence of plasmid-mediated quinolone resistance genes qnr and aac(6′)-Ib-cr in clinical isolates of Enterobacteriaceae from nine teaching hospitals in China. *Antimicrob. Agents Chemother.* 52, 4268–4273. <https://doi.org/10.1128/AAC.00830-08>.
- Yuan, J., Xu, X., Guo, Q., Zhao, X., 2012. Prevalence of the oqxAB gene complex in *Klebsiella pneumoniae* and *Escherichia coli* clinical isolates. *J. Antimicrob. Chemother.* 67, 1655–1659. <https://doi.org/10.1093/jac/dks086>.
- Zhang, X.X., Zhang, T., Fang, H.H.P., 2009. Antibiotic resistance genes in water environment. *Appl. Microbiol. Biotechnol.* 82, 397–414. <https://doi.org/10.1007/s00253-008-1829-z>.
- Zhang, C., Cui, F., Zeng, G.M., Jiang, Z., Yang, Z., Yu, M., Zhu, Y., Shen, L., 2015. Quaternary ammonium compounds (QACs): a review on occurrence, fate and toxicity in the environment. *Sci. Total Environ.* 518, 352–362. <https://doi.org/10.1016/j.scitotenv.2015.03.007>.
- Zou, L., Meng, J., McDermott, P.F., Wang, F., Yang, Q., Cao, G., Hoffmann, M., Zhao, S., 2014. Presence of disinfectant resistance genes in *Escherichia coli* isolated from retail meats

- in the USA. *J. Antimicrob. Chemother.* 69, 2644–2649. [https://doi.org/10.1093/jac/](https://doi.org/10.1093/jac/dku197)
[dku197](https://doi.org/10.1093/jac/dku197).
- Zulkeflee, S.N.M., Yusaimi, Y.A., Sugiura, N., Iwamoto, K., Goto, M., Utsumi, M., Othman, N.B., Zakaria, Z., Hara, H., 2016. Phenotypic and genetic characterization of multidrug-resistant *Staphylococcus aureus* in the tropics of Southeast Asia. *Microbiology* 162, 2064–2074. <https://doi.org/10.1099/mic.0.000392>.
- Zurfluh, K., Abgottspon, H., Hachler, H., Nuesch-Inderbinen, M., Stephan, R., 2014. Quinolone resistance mechanisms among extended spectrum beta lactamase (ESBL) producing *Escherichia coli* isolated from rivers and lakes in Switzerland. *PLoS One* 9, e95864. <https://doi.org/10.1371/journal.pone.0095864>.